
Faculty of Engineering and Physical Sciences
Department of Civil, Maritime and Environmental Engineering

**Digital-Twin-Enabled Life-Cycle Assessment of
Sustainable Pavement Materials for Nigerian Roads**

FEEG3003 Individual Project

Academic Year 2025–2026

Name: Faizaan Umar

Student ID: 34411364

Supervisor: Prof Paul Kemp

Word count: 10,200

Date submitted: 01/05/2026

This dissertation is submitted in partial fulfilment of the requirements for the degree of Bachelor of Engineering (with Honours), Faculty of Engineering and Physical Sciences, University of Southampton.

I confirm that this dissertation is my own work. Where the work of others has been used, it has been explicitly acknowledged and referenced. I have read and understood the University's policy on academic integrity and plagiarism.

Declaration

I, Faizaan Umar, declare that this thesis and the work presented in it are my own and has been generated by me as the result of my own original research.

I confirm that:

1. This work was done wholly or mainly while in candidature for a degree at this University.
2. Where any part of this thesis has previously been submitted for any other qualification at this University or any other institution, this has been clearly stated.
3. Where I have consulted the published work of others, this is always clearly attributed.
4. Where I have quoted from the work of others, the source is always given. With the exception of such quotations, this thesis is entirely my own work.
5. I have acknowledged all main sources of help.
6. Where the thesis is based on work done by myself jointly with others, I have made clear exactly what was done by others and what I have contributed myself.
7. None of this work has been published before submission.

Acknowledgements

First and foremost, all praise and gratitude belongs to Allah, the Most Merciful, the Most Kind. Every ability I have, every opportunity I have been given, and every moment of clarity during this work came from Him alone. This dissertation, like everything else, begins and ends with His grace.

To my parents, none of this exists without you. Dad, your discipline, your sacrifices, and the quiet way you have always believed in me have shaped everything I am. Mum, your warmth, your prayers, and your unconditional love have carried me through every difficult moment far from home. You both gave up more than I will ever fully understand so that I could have opportunities you never had. I hope this is a small return on that investment.

To my siblings, Rayyan, Aariz, and Amana. Growing up with you three has been the greatest constant in my life. Rayyan and Aariz, watching you both grow into your own people has pushed me to be someone worth looking up to. Amana, your energy and spirit remind me what actually matters. I am proud of all three of you every single day.

To my supervisor, Professor Paul Kemp, thank you for your guidance, patience, and honest feedback throughout this project. Your expertise, patience, and mentorship throughout this process have been invaluable. Thank you for pushing me to achieve my best and for guiding this research to its completion.

To Vinoth, Nana, Richard, Tunmise, Josh, AbdulRahman, Abdullahi, Baran, and Ibrahim, my engineering family at Southampton. The late nights, the stressed B25/library sessions, the disagreements about methods, and the moments of relief when things finally worked: thank you for making the hard parts easier and the good parts worth celebrating. I couldn't have asked for a better group to walk this path with.

To my friends who have been part of this journey from long before Southampton, and to those I found here who made Southampton feel like home, you know who you are. Whether we met recently or have known each other for years, your friendship has been a vital escape and a constant support system. The calls, the check-ins, the distractions when I needed them and the motivation when I did not: thank you.

To everyone mentioned above: thank you for being part of my story. Your support, in all its various forms, has been the fuel that kept me moving forward through every high and low. I am truly blessed to have you all in my life.

Abstract

This study develops a two-layer computational framework for evaluating pavement materials under Nigerian road conditions. An asset-level MATLAB simulation models deterioration, maintenance, cost, and CO₂ for conventional asphalt, sealed laterite, and plastic-modified asphalt over 20 years; outputs are coupled one-way into a Vensim system dynamics model of a 100 km network tested under seven policy scenarios. At asset level, sealed laterite has the lowest cost (NPV £79,688) and construction-phase CO₂ (44.7 t). A condition-dependent rolling-resistance model narrows laterite's whole-life CO₂ advantage from 151 t to 99 t over asphalt. At network level, a critical inversion emerges: laterite's short service life generates deterioration faster than rehabilitation can clear it, making it the worst network scenario despite being the best asset-level material. Budget variation of $\pm 30\%$ produces no backlog change; the binding constraint is deterioration rate, not expenditure. Overload enforcement is the only lever simultaneously reducing backlog, cost, and CO₂, with WIM station payback of 1.4–3.8 years. The combined enforcement-and-plastic scenario ranks first across 83% of multi-criteria weight samples and all 1,000 Monte Carlo draws. The principal finding is that Nigerian road network performance is constrained by how fast roads deteriorate, not by how much is spent on rehabilitation.

Contents

Declaration	2
Acknowledgements	3
Abstract	1
List of Figures	6
List of Tables	8
Abbreviations and Nomenclature	9
1 Introduction	12
1.1 Background and Context	12
1.2 Problem Statement and Research Gap	12
1.3 Aim, Objectives and Research Questions	12
1.4 Scope and Limitations	13
1.5 Dissertation Structure	13
2 Literature Review	14
2.1 Nigerian Road Infrastructure and Pavement Performance	14
2.2 Pavement Materials and Sustainable Alternatives	14
2.3 Pavement LCA and LCCA	15
2.4 Digital Twins for Pavement and Road Infrastructure	15
2.5 Systems Thinking and System Dynamics in Road Networks	16
2.6 Synthesis and Research Gap	16
3 Methodology	18
3.1 Study Design and Analytical Scope	18
3.2 Asset-Level Digital Twin in MATLAB	21
3.2.1 Model Purpose and Initial Condition	21
3.2.2 Traffic Loading	21
3.2.3 Deterioration Model	21
3.2.4 Maintenance Triggers and Interventions	22
3.2.5 Cost and CO ₂ Accounting	22
3.2.6 Implementation	23
3.3 Network-Level System Dynamics Model in Vensim	24
3.3.1 Stock-Flow Structure and Initial Conditions	24
3.3.2 Rehabilitation and Budget Logic	24
3.3.3 Coupling with the Digital Twin	26
3.3.4 Cost and Emissions Accumulation	27

3.4	Scenario Design	28
3.5	Multi-Criteria Decision Analysis	28
3.5.1	Weight Sampling and Robustness	29
3.5.2	Budget-Equivalent Framing of Enforcement Benefits	29
3.6	Model Verification, Plausibility Checks, and Sensitivity Analysis	29
3.7	Limitations	32
4	Results	33
4.1	Asset-Level Material Comparison	33
4.1.1	Intervention Frequency and Calibration Validation	33
4.1.2	Condition Index Trajectories	33
4.1.3	Life-Cycle Cost and CO ₂	33
4.1.4	25-Year Horizon Sensitivity	35
4.1.5	Monte Carlo Uncertainty Analysis	35
4.1.6	Sobol Variance-Based Sensitivity Analysis	36
4.2	Network-Level Scenario Analysis	38
4.2.1	Backlog Trajectories and the Non-Binding Budget Result	38
4.2.2	Initial Backlog Sensitivity	40
4.3	Priority-Based Allocation Sensitivity	40
4.4	Network-Level Monte Carlo	42
4.5	MCDA Robustness Check	43
4.6	Budget-Equivalent Analysis of Enforcement Benefits	43
5	Discussion	45
5.1	Asset-Level Material Performance (RQ1)	45
5.2	Network-Level Findings and Their Interpretation	45
5.3	Policy Implications	46
5.4	Limitations and Contextualisation	48
6	Conclusions and Recommendations	49
6.1	Summary of Findings	49
6.2	Policy Recommendations	49
6.3	Decision Framework for Material Selection	50
6.4	Research Contributions Summary	50
6.5	Limitations and Future Work	51
6.6	Concluding Statement	51
A	Parameter Register	56
A.1	General Model Settings	56
A.2	Traffic, Climate, and Condition Inputs	57
A.3	Deterioration Structure	57
A.4	Maintenance and Intervention Logic	58
A.5	Cost Parameters	58
A.6	Carbon Parameters	59
A.7	Calibration Targets	59
A.8	Exported MATLAB-to-Vensim Parameters	59

A.9	Use-Phase CO ₂ Parameters	60
A.10	Notes	61
B	Key MATLAB Code Listings	62
B.1	rrc_condition_factor.m: Condition-Dependent RRC Scaling	62
B.2	config_base.m: Baseline Parameter Configuration	63
B.3	simulate_section.m: Annual Time-Step Loop	66
B.4	deterioration_step.m: Annual Condition Loss	68
B.5	event_cost_emissions.m: Intervention Cost and CO ₂	69
C	Vensim Model Equations and Coupling CSV	72
C.1	Purpose of the Vensim Model	72
C.2	Full Vensim Equation Listing	72
C.3	Additional Equations: Use-Phase CO ₂ and Priority Allocation	75
C.4	Exported MATLAB-to-Vensim Coupling CSV	76
C.5	Commentary	76
D	Supplementary Sensitivity and Monte Carlo Plots	77
D.1	Initial Backlog Sensitivity	77
D.2	One-at-a-Time Sensitivity Analysis	77
D.3	Monte Carlo Box Plots	77
D.4	Appendix Summary	78
E	Structural Adequacy Check	79
E.1	Purpose and Scope	79
E.2	Design Traffic Estimate	79
E.3	Thickness Comparison Against ORN 31	79
E.4	ESA Sensitivity Check	80
E.5	Conclusion	80

List of Figures

2.1	Research positioning matrix. Studies mapped by modelling scale and assessment scope; the network-level, multi-metric, material-specific quadrant identifies the gap this study addresses.	17
3.1	CI-dependent RRC mechanism. (a) The $\phi(\text{CI})$ scaling function (Eq. (7)); dashed lines show each material's mean in-service CI. (b) Cumulative use-phase CO ₂ trajectories; laterite overtakes asphalt because its shorter TTL sustains a lower mean CI, amplifying rolling-resistance emissions despite having the same baseline RRC.	23
3.2	Digital-twin-enabled framework workflow: inputs flow through deterioration, condition update, and threshold-triggered intervention to produce asset-level outputs exported to Vensim.	24
3.3	System dynamics stock-flow structure: material-specific acceptable-condition stocks deteriorate into backlog; rehabilitation draws from backlog subject to budget constraints; cost and CO ₂ accumulate separately.	25
4.1	Condition index trajectories under baseline traffic and climate conditions (AADT = 13,667, $k_{\text{climate}} = 1.15$, ESA/HGV = 10.0).	34
4.2	Whole-life CO ₂ by phase. Solid bars: construction-phase; hatched bars: use-phase rolling-resistance (CI-dependent RRC model). Percentages show each phase's share of the whole-life total. Asphalt's 58% construction-phase share contradicts the use-phase-dominated pattern reported by Santero et al. (2011a) for temperate-climate studies.	34
4.3	Cumulative undiscounted agency cost (left) and embodied CO ₂ (right) for all three materials over the 20-year horizon.	35
4.4	Monte Carlo uncertainty distributions ($N = 2,000$) for NPV cost, construction-phase CO ₂ , and first overlay year across six sampled parameters (ODM and initial backlog varied separately in the network-level Monte Carlo, Table 4.10).	36
4.5	One-at-a-time sensitivity analysis: percentage change in NPV cost from baseline. Compare with Table 4.4: OAT correctly identifies the dominant NPV drivers but understates climate multiplier's role in CO ₂ uncertainty, which operates almost entirely through interaction with α	38
4.6	Sobol sensitivity analysis: first-order (S_1 , blue) and total-order (S_T , red) indices for NPV and whole-life CO ₂ for each material. Parameters with $S_T \gg S_1$ act primarily through interactions. The $k_{\text{clim}}-\alpha$ interaction dominates CO ₂ uncertainty for asphalt and plastic; $f_{\text{mat, laterite}}$ drives laterite CO ₂ directly.	38
4.7	Network backlog trajectories for all seven scenarios. S0, S1, and S2 overlap completely, confirming the budget is non-binding; the binding constraint is deterioration inflow.	39

4.8	MCDA rank probabilities under a wide uniform weight space ($w_i \geq 0.05$, $\sum w_i = 1$, $N = 50,000$ Dirichlet samples). S6 ranks first in 82.9% of samples; S5 ranks first in 14.4%; S4 in 2.7% (when CO ₂ or cost weight dominates). S0, S1, S2, and S3 never achieve rank 1 across the sampled space. As a boundary check, the analysis was also run with each weight constrained to a narrow band around its Dirichlet mean: S6 retained first rank in all cases, confirming the ordering is not sensitive to the shape of the weight distribution.	43
4.9	Annual budget-equivalent of enforcement (left) and cumulative co-benefits vs S0 baseline (right). CO ₂ savings shown are construction-phase only; whole-life savings are approximately 6% (S5) and 9.7% (S6).	44
5.1	Network scenario performance matrix. Each cell shows the raw metric value; colour indicates relative performance across all seven scenarios (green = best, red = worst, normalised within each metric). S6 (outlined) achieves the best or near-best outcome on all four metrics. S0, S1 and S2 are identical, confirming the budget is non-binding. S4's green CO ₂ cells alongside its red backlog cell show the trade-off that makes asset-level LCA an insufficient basis for network strategy.	47
D.1	Supplementary one-at-a-time sensitivity tornado chart showing percentage change in 20-year NPV cost relative to baseline for the tested parameter variations across the three pavement materials.	77
D.2	Supplementary Monte Carlo box plots ($N = 2,000$) for 20-year NPV cost, construction-phase CO ₂ , and first overlay year. Bounds reflect six sampled parameters; ODM, initial backlog fraction, and rehabilitation shares were not varied at the asset level. ODM and initial backlog are varied in the network-level Monte Carlo (Table 4.10).	78

List of Tables

2.1	Material comparison summary. Sources: Xu et al. (2022); Jwaida et al. (2023); Pan et al. (2025); Ugwuanyi and Onyelowe (2019); Alade and Amu (2025); Pinard et al. (2003); Kenya Ministry of Transport, Infrastructure, Housing and Urban Development (2017).	15
3.1	Baseline parameter values and sensitivity ranges.	19
3.1	Baseline parameter values and sensitivity ranges (<i>continued</i>).	20
3.2	MATLAB → Vensim one-way coupling mapping.	27
3.3	Scenario matrix.	28
3.4	Summary of principal validation and plausibility checks.	30
4.1	Intervention frequency and life-cycle outcomes: baseline simulation.	33
4.2	Horizon sensitivity: 20-year vs 25-year outcomes (construction-phase CO ₂ and NPV cost).	35
4.3	Monte Carlo summary statistics ($N = 2,000$, seed = 42).	36
4.4	Sobol sensitivity indices. S_1 = first-order; S_T = total-order (includes interactions). $N = 1,024$ base samples; Jansen estimator (Saltelli et al., 2010). Values below 0.01 shown as —. High $S_T - S_1$ = primarily interactive.	37
4.5	Network-level scenario outcomes. Con. CO ₂ = construction-phase; W-L CO ₂ = whole-life.	39
4.6	Budget threshold analysis (selected factors).	40
4.7	Priority-based allocation sensitivity: year-20 backlog (km). Positive Δ = priority allocation is worse.	41
4.8	TTL laterite sensitivity: inversion persists to TTL 4.5 yr, breakeven at 5.0 yr.	41
4.9	Value-of-information analysis for laterite TTL. EVPI = maximum rational expenditure on corridor-specific TTL measurement. Cost of error at calibrated TTL: selecting S4 over S0 when TTL = 3.33 yr costs £43,672 in additional backlog rehabilitation (£437/km of network).	42
4.10	Network-level Monte Carlo: year-20 backlog distribution (km) for S0, S5, S6 across $N = 1,000$ LHS samples. $S6 < S5 < S0$ in all 1,000 samples.	42
4.11	Budget-equivalent interpretation of enforcement benefits.	43
6.1	Material selection decision framework. TTL evidence = corridor-specific laterite service-life data meeting or exceeding the asphalt benchmark (≥ 5 yr); CO ₂ priority = CO ₂ reduction carries direct procurement or policy weight; Low discount rate = below $\approx 8\%$. Recommendations are conditional on the modelled parameterisation; agencies should adjust thresholds using local data.	50
6.2	Research questions, key findings, and policy implications.	51
A.1	General model settings from <code>config_used.json</code>	56
A.2	Traffic, climate, and condition inputs from <code>config_used.json</code>	57

A.3	Deterioration parameters from <code>config_used.json</code>	57
A.4	Maintenance logic from <code>config_used.json</code>	58
A.5	Cost parameters from <code>config_used.json</code>	58
A.6	Material-specific intervention cost values from <code>config_used.json</code>	58
A.7	Carbon and material-emissions parameters from <code>config_used.json</code>	59
A.8	Calibration target windows from <code>config_used.json</code>	59
A.9	Material-specific parameters exported to <code>vensim_params.csv</code>	59
A.10	Additional exported material results from <code>vensim_params.csv</code>	60
A.11	Use-phase rolling-resistance CO ₂ parameters. Sources: Trupia et al. (2017); Chatti & Zaabar (2012) NCHRP 720; IPCC (2006) Tier 1.	60
A.12	Asset-level whole-life CO ₂ summary. Construction-phase values from <code>vensim_params.csv</code> ; use-phase values computed from Eq. (6) with condition-index-dependent RRC scaling (Eq. (7)), 4%/yr traffic growth.	60
C.1	Material-specific parameters exported to <code>vensim_params.csv</code>	76
C.2	Maintenance interval and annualised parameters from <code>vensim_params.csv</code>	76
C.3	Lifetime cost, carbon, and intervention counts from <code>vensim_params.csv</code>	76
D.1	Initial backlog sensitivity: year-20 backlog (km) for S0, S1, S2 and selected scenarios under varying initial backlog fractions. S6 outperforms S5 and S4 is worse than S0 at all levels tested.	77
E.1	Design traffic estimate for structural adequacy check.	79
E.2	Thickness comparison: modelled section vs ORN 31 reference (CBR = 5%, T5 traffic class).	80
E.3	ESA sensitivity check: 20-year cumulative traffic and ORN 31 class under alternative assumptions.	80

Abbreviations and Nomenclature

Abbreviations

Abbreviation	Definition
AADT	Annual Average Daily Traffic
CBR	California Bearing Ratio
CI	Condition Index (pavement condition scale 0–100)
CO ₂	Carbon dioxide (sole environmental metric used in this study)
ESA	Equivalent Standard Axle
EVPI	Expected Value of Perfect Information
GBP	Pounds Sterling
GHG	Greenhouse Gas
HGV	Heavy Goods Vehicle
IRI	International Roughness Index (m/km)
LCA	Life-Cycle Assessment
LCCA	Life-Cycle Cost Analysis
LHS	Latin Hypercube Sampling
MATLAB	Matrix Laboratory (MathWorks simulation software)
MCDA	Multi-Criteria Decision Analysis
NGN	Nigerian Naira
NPV	Net Present Value
OAT	One-At-a-Time (sensitivity analysis method)
ODM	Overload Damage Multiplier
ORN	Overseas Road Note (TRL)
PSR	Present Serviceability Rating (pavement condition scale 0–5)
RQ	Research Question
RRC	Rolling Resistance Coefficient (dimensionless)
SD	System Dynamics
TRL	Transport Research Laboratory
TTL	Time-to-deterioration proxy used in the network model (years)
VoI	Value of Information
WIM	Weigh-in-Motion (dynamic axle-load measurement system)

Mathematical Symbols

Symbol	Definition
<i>General and asset-level model</i>	
T	Analysis horizon (years)

continued on next page

(Mathematical Symbols, continued)

Symbol	Definition
t	Time index (years)
Δt	Time step (1 year)
d	Annual discount rate
g	Annual traffic growth rate
α	Loading exponent in traffic damage scaling
CI_t	Condition Index at year t (0–100 scale)
ΔCI_t	Annual condition-index loss at year t
R_t	Condition recovery from maintenance intervention at year t
r_b	Baseline deterioration rate (CI points/year); piecewise by condition band
$r_{\text{good}}, r_{\text{mid}}, r_{\text{poor}}, r_{\text{vpoor}}$	Band-specific deterioration rates (CI points/year)
$f_{\text{traffic},t}$	Traffic loading factor at year t
f_{climate}	Climate multiplier for humid tropical exposure
k_{clim}	Numerical value of the climate multiplier
f_{material}	Material-specific deterioration factor
f_{mat}	Shorthand for f_{material} (subscripted by material)
ESA_t	Annual equivalent standard axles at year t
ESA_{ref}	Reference annual ESAs (non-overloaded baseline)
$AADT_0$	Initial two-way annual average daily traffic (vehicles/day)
p_{HGV}	Heavy goods vehicle share of total traffic
C_t	Agency cost incurred at year t (GBP)
E_{con}	Cumulative construction-phase CO ₂ (tonnes)
E_t	Construction-phase CO ₂ emitted at year t (tonnes)
$E_{\text{use},t}$	Use-phase rolling-resistance CO ₂ at year t (tonnes)
E_{wl}	Whole-life CO ₂ (construction + use phase, tonnes)
<i>Rolling-resistance CO₂ model</i>	
$\phi(\text{CI})$	CI-dependent RRC scaling factor (dimensionless; ≥ 1)
β	Mixed-fleet roughness–fuel sensitivity coefficient (0.014)
RRC_m	Rolling resistance coefficient for material m (dimensionless)
F_{mix}	Fleet-average fuel factor (L/veh-km per unit RRC)
VKT_{yr}	Annual vehicle-kilometres per kilometre of road
EF	CO ₂ emission factor for diesel (2.64 kg CO ₂ /L)
λ	Lane split factor (0.5 for one design lane)
L	Road section length (km)
<i>Network-level system dynamics model</i>	
A_m	Acceptable-condition stock for material m (km)
m	Material index (asphalt, laterite, plastic)
s_m	Rehabilitation share for material m (fixed proportion)
c_m	Unit rehabilitation cost for material m (GBP/km)

continued on next page

(Mathematical Symbols, continued)

Symbol	Definition
e_m	Unit rehabilitation CO ₂ for material m (t/km)
TTL_m	Time-to-backlog (mean service life) for material m (years)
TTL_m^{eff}	Effective TTL after overload damage adjustment (years)
τ_{rehab}	Rehabilitation response time (1 year; dimensional fix)
τ_m	TTL for material m used in priority-allocation equations
π_m	Priority weight for material m in priority-based allocation
$\dot{D}_m$	Deterioration flow for material m (km/year)
<i>Multi-criteria decision analysis</i>	
S_i	Composite MCDA score for scenario i
w_1, w_2, w_3	Weights for cost, CO ₂ , and backlog criteria respectively
$\tilde{x}_{ik}$	Min-max normalised value of criterion k for scenario i
<i>Sobol sensitivity analysis</i>	
S_1	First-order Sobol sensitivity index (direct effect)
S_T	Total-order Sobol sensitivity index (direct + interaction effects)

1 Introduction

1.1 Background and Context

Road infrastructure in Nigeria suffers from persistent deterioration, maintenance backlogs, and repeated rehabilitation cycles driven by weak lateritic subgrades, inadequate drainage, axle overloading, and insufficient geotechnical characterisation (Foster and Pushak, 2011; Yakubu et al., 2023). These conditions make the Nigerian context particularly suited to life-cycle evaluation rather than initial-cost comparison alone.

Sealed laterite offers local-material and cost advantages; plastic-modified asphalt may improve rutting resistance but long-term durability is uncertain (Xu et al., 2022; Jwaida et al., 2023). Life-cycle assessment (LCA) and life-cycle cost analysis (LCCA) provide the evaluation framework (ISO, 2006a,b; Santero et al., 2011a), while digital-twin and system dynamics (SD) approaches can link deterioration to policy outcomes but remain rarely integrated in low-data settings (Talaghat et al., 2024; Linard, 2009).

1.2 Problem Statement and Research Gap

Three gaps motivate this study. First, pavement LCA studies rely on assumed service lives and fixed maintenance schedules rather than condition-driven intervention timing (Santero et al., 2011a), meaning material rankings may depend on the schedule assumption rather than the material. Second, pavement digital twins are designed for sensor-rich environments, limiting transfer to data-constrained agencies. Third, no integrated framework combines material-specific pavement simulation, life-cycle cost and carbon assessment, and network-level SD analysis for Sub-Saharan African conditions. These three gaps interact: without condition-driven scheduling, LCA cannot detect the maintenance-frequency effects that only become visible at network scale.

1.3 Aim, Objectives and Research Questions

The aim of this study is to develop a digital-twin-enabled, scenario-based framework for evaluating the long-term condition, cost, and carbon performance of alternative pavement materials for Nigerian roads.

The objectives are:

1. To review literature on Nigerian pavement performance, sustainable materials, LCA/LCCA, digital twins, and system dynamics.
2. To develop a MATLAB-based low-data asset simulation modelling deterioration, interventions, cost, and CO₂ for asphalt, sealed laterite, and plastic-modified asphalt.
3. To construct a Vensim SD model of network-level backlog, cost, and emissions and couple it to the asset model.
4. To test budget, adoption-mix, and enforcement scenarios over 20 years.
5. To generate evidence-based recommendations for sustainable pavement strategy selection in

Nigeria.

The study addresses three research questions:

- **RQ1:** What are the 20-year life-cycle cost, CO₂, and condition trajectories for each material at the asset level?
- **RQ2:** How does one-way coupling of asset-level outputs into a system dynamics model enable network-level interpretation of material performance and policy effectiveness?
- **RQ3:** How do budget level, adoption mix, and overload enforcement affect network backlog, cost, and emissions over 20 years?

The dissertation makes three contributions: (i) a condition-driven low-data asset model generating endogenous maintenance schedules; (ii) a one-way coupled network SD model enabling asset-to-network interpretation of backlog, cost, and carbon; and (iii) an integrated decision-support framework comparing condition, cost, and carbon for Nigerian pavement strategy.

1.4 Scope and Limitations

The functional unit is 1 km of single-lane flexible pavement over 20 years under humid tropical Nigerian conditions. Environmental scope covers construction-phase and use-phase rolling-resistance CO₂ only; end-of-life effects, toxicity, and microplastics are excluded. Cost scope covers agency costs only. The SD model represents a stylised 100 km network.

1.5 Dissertation Structure

Chapter 2 reviews the five literature strands underpinning the study. Chapter 3 presents the asset model, SD model, coupling strategy, and scenario design. Chapter 4 reports asset- and network-level results. Chapter 5 discusses findings against the research questions. Chapter 6 presents conclusions and recommendations.

2 Literature Review

2.1 Nigerian Road Infrastructure and Pavement Performance

Foster and Pushak (2011) show that Nigerian road maintenance has been structurally underfunded, with only 67% of paved roads in good or fair condition. Geotechnical case studies consistently associate premature failure with water ingress, inadequate drainage, and weak subgrades rather than traffic loading alone.

A critical reading of the Nigerian literature shows convergence on failure mechanisms, but not on one uniform geology. Daramola et al. (2018) focus on mineralogical variability in south-western soils, identifying clay-mineral composition as the primary moisture-sensitivity driver. Akudo and Aigbadon (2024) reach a similar conclusion in north-central Nigeria but emphasise bearing capacity rather than mineralogy, while Ademila (2017) use comparative geotechnical profiling of failed and stable sections in the south-west to show that failure thresholds vary even within the same corridor. The three studies converge on drainage, moisture sensitivity, and subgrade quality as dominant controls, but the mechanism differs by geology, an important caution against applying any single corridor's deterioration rates to the national level.

Yakubu et al. (2023) synthesise these failure mechanisms, but no Nigerian study reports quantitative deterioration rates usable for parametric modelling. The deterioration parameters in this study are therefore calibrated to literature targets rather than extracted from condition surveys (Sections 3.6 and 5.4). Regional variability in lateritic materials further cautions against transferring any single corridor's parameters nationally.

2.2 Pavement Materials and Sustainable Alternatives

Conventional asphalt remains the practical baseline for pavement comparison because its design methods, supply chains, and maintenance practices are well established. Asphalt performance in humid tropical environments is shaped by high temperatures, moisture ingress, and drainage quality in addition to traffic loading (Transport Research Laboratory, 1993; Pinard et al., 2003), meaning that service-life assumptions from temperate environments cannot be transferred to Nigerian conditions without caution. Asphalt therefore remains the most defensible benchmark, but not necessarily the automatically superior option in whole-life terms.

For plastic-modified asphalt, Xu et al. (2022) and Jwaida et al. (2023) confirm improved rutting resistance but note dependence on dosage, pre-treatment, and unresolved ageing concerns. Pan et al. (2025) foreground toxic emissions and microplastics. Reported CO₂ advantages are typically construction-phase only (Trupia et al., 2017). Neither source characterises performance under sustained tropical exposure; durability benefits may be optimistic for Nigerian conditions.

Ugwuanyi and Onyelowe (2019) and Alade and Amu (2025) confirm Nigerian lateritic soils are abundant but variable, with borrow-pit quality ranging from subbase-acceptable to subgrade-only. The SADC guideline (Pinard et al., 2003) and Kenya manual (Kenya Ministry of Transport, Infrastructure, Housing and Urban Development, 2017) report seal lives of 5–12 years for low-cost seals

over laterite bases. However, both calibrate to low-volume roads (<300 vehicles/day), roughly 45 times less traffic than the 13,667 vehicles/day baseline used here. Higher traffic amplifies deterioration through the fourth-power law, so the 3.33 yr TTL adopted in this study is more likely an upper bound on federal-corridor laterite performance than a lower bound. The laterite inversion finding is therefore robust to this calibration uncertainty.

Table 2.1: Material comparison summary. Sources: Xu et al. (2022); Jwaida et al. (2023); Pan et al. (2025); Ugwuanyi and Onyelowe (2019); Alade and Amu (2025); Pinard et al. (2003); Kenya Ministry of Transport, Infrastructure, Housing and Urban Development (2017).

Material	Performance evidence	LCA/LCCA signal	Main uncertainty
Conventional asphalt	Strong baseline performance, familiar maintenance practice	Mature evidence base but sensitive to maintenance and boundary assumptions	Tropical climate transferability, use-phase and maintenance assumptions
Plastic-modified asphalt	Often improved rutting/high-temperature stability; variable ageing and compatibility	Can reduce impacts, but only under favourable recycling and processing assumptions	Toxic emissions, microplastics, end-of-life allocations
Sealed laterite	Low-cost and local where material quality and drainage are adequate	Economic evidence stronger than formal LCA evidence	Moisture sensitivity, borrow-pit variability, reseal discipline

2.3 Pavement LCA and LCCA

Pavement LCA is methodologically mature but not harmonised (ISO, 2006a,b; Santero et al., 2011a; AzariJafari et al., 2016; Harvey et al., 2016). The most contested area is the use phase. Santero et al. (2011a) identify use-phase as the principal uncertainty source, implying material selection should prioritise surface texture. Trupia et al. (2017) partially support this, but their RRC parameters derive from US interstate conditions (IRI 1–3 m/km). On Nigerian federal roads, maintained sections exhibit IRI of 4–8 m/km (Oyekanmi et al., 2020), meaning construction frequency rather than surface texture drives roughness. The two studies imply opposite data-collection priorities for tropical contexts, and neither resolves the question.

Noshadravan et al. (2013) and Gregory et al. (2016) show that probabilistic LCA can make section-level material rankings robust without network-level analysis. This study’s Monte Carlo confirms that laterite’s 95th-percentile CO₂ (88.9 t) falls below the 5th percentile for plastic (109.0 t), satisfying the Noshadravan criterion. However, this answers “which material has lower section-level impact?” but not “does adopting it at scale change backlog dynamics?” The laterite inversion demonstrates precisely this gap: section-level robustness does not predict network-level performance.

2.4 Digital Twins for Pavement and Road Infrastructure

Kritzinger et al. (2018) distinguish digital models, digital shadows, and full digital twins by degree of data exchange. Pavement DT research is dominated by sensor-rich architectures (Talaghat et al., 2024; Bello et al., 2022; Liu et al., 2023) that assume continuous data environments most Sub-Saharan African agencies cannot sustain, and rarely integrates cost or carbon as core state variables. In this dissertation, *digital-twin-enabled* denotes a simulation structured to accept future data feeds rather than a claim of live bidirectional sensing.

2.5 Systems Thinking and System Dynamics in Road Networks

Linard (2009, 2010) demonstrate how budget constraints and roughness produce backlog behaviour that conventional pavement-management systems obscure. Ruiz and Guevara (2020a,b) show that network condition, cost, and emissions co-evolve in Chilean and Colombian applications. Shiboub and Assaf (2022) extend this to network-level rehabilitation. The shared limitation is that deterioration is parameterised as a single aggregate rate rather than by surfacing material, so none can answer whether substituting laterite or plastic changes network dynamics.

Across these studies, road condition is a stock, deterioration an outflow, and maintenance a restorative counter-flow. What remains missing is not SD itself but the input structure: existing models rely on aggregate deterioration, not coupling from material-specific simulations.

2.6 Synthesis and Research Gap

The literature establishes four robust propositions. First, Nigerian road deterioration is driven by a recurring interaction of weak preventive maintenance, poor drainage, and variable subgrade quality. Second, sustainable material alternatives exist, but the evidence is uneven. Third, pavement LCA/LCCA is well established in principle but remains weakened by inconsistent boundaries, static maintenance assumptions, and weak region-specific inventories. Fourth, digital-twin-enabled and system dynamics approaches are both promising, yet the former remains sensor-centric and the latter usually operates at too aggregated a level to compare material pathways rigorously.

These gaps are not independent: dominant DT frameworks assume sensor infrastructures many agencies lack, while existing SD models are too aggregate to propagate material-specific outcomes into network-level cost and carbon trajectories. No single literature strand, taken alone, answers the decision problem facing a Nigerian road agency.

The critical insight is that these limitations are structurally interdependent rather than independent. Static LCA is blind to deterioration dynamics because it assumes away the feedback between maintenance frequency and service-life duration; SD models are blind to material heterogeneity because they aggregate deterioration to a single rate. Neither limitation can be fixed by adding more data to the other approach. A model that runs SD with aggregate deterioration cannot detect the laterite inversion; a model that runs LCA with condition-driven scheduling cannot detect the budget non-binding result. The gap is architectural, not merely parametric, and this is the structural motivation for the two-layer framework: not to improve either model in isolation, but to make each visible to the blind spot of the other.

Accordingly, the methodological challenge is to construct a framework that preserves material-specific deterioration behaviour at asset level while enabling policy-scale interpretation of backlog, cost, and carbon at network level, under the realistic data and institutional constraints of Sub-Saharan African road agencies.

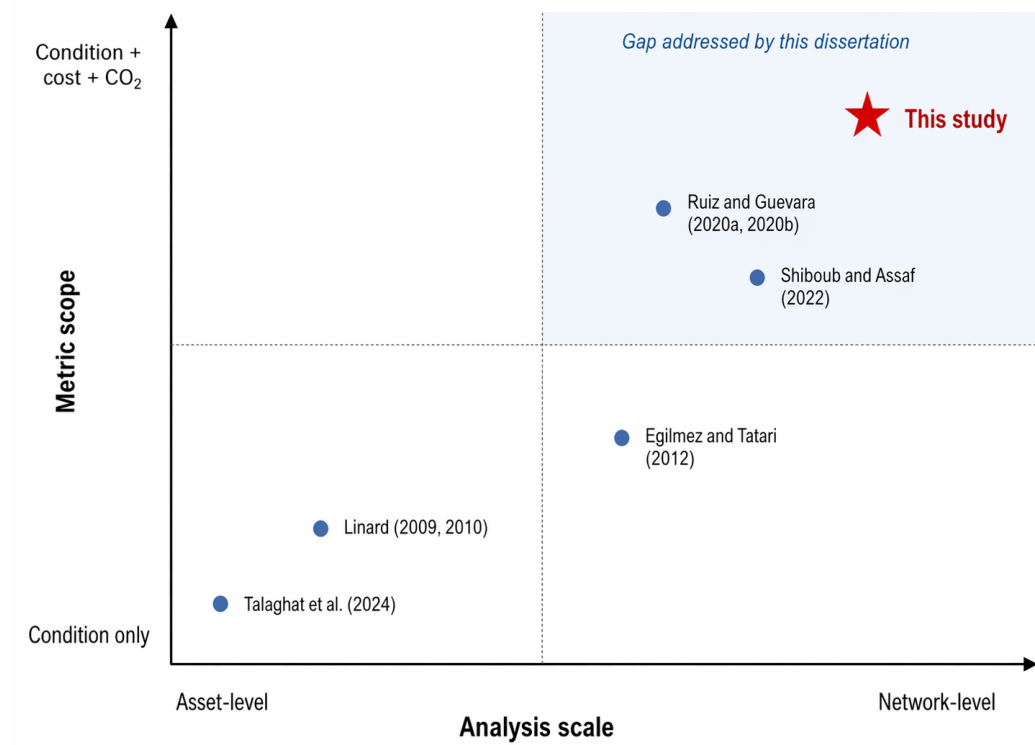


Figure 2.1: Research positioning matrix. Studies mapped by modelling scale and assessment scope; the network-level, multi-metric, material-specific quadrant identifies the gap this study addresses.

3 Methodology

3.1 Study Design and Analytical Scope

This study adopts a two-layer decision-support modelling framework. The first layer is an asset-level MATLAB simulation modelling deterioration, maintenance interventions, agency cost, and CO₂ for a representative flexible pavement section. The second layer is a Vensim SD model translating asset-level outputs into network-scale behaviour under varying scenarios. The framework is sequential and one-way coupled: MATLAB generates material-specific outputs transferred to the SD model as parameters. The asset model addresses RQ1, the coupling architecture addresses RQ2, and the SD scenario analysis addresses RQ3.

The functional unit is 1 km of single-lane flexible pavement (lane width 3.65 m). Traffic parameters are from the Lokoja–Abuja corridor (Oyekanmi et al., 2020); deterioration parameters reflect southern Nigerian conditions (Pinard et al., 2003). The analysis horizon is 20 years with annual time steps; costs are in NGN converted to GBP at 1,024.4 NGN/£. Baseline assumptions are in Table 3.1, coupling in Table 3.2, and scenarios in Table 3.3.

Table 3.1: Baseline parameter values and sensitivity ranges.

(a) General model parameters				
Parameter	Symbol	Baseline	Range	Source / basis
Analysis horizon	T	20 yr	Fixed	Project scope
Time step	Δt	1 yr	Fixed	Annual planning
Functional unit	—	1 km, 1 lane, 3.65 m	Fixed	Project scope
Discount rate	d	0.12	0.08–0.15	World Bank guidance
AADT (two-way)	AADT ₀	13,667 veh/day	Fixed	Lokoja–Abuja corridor: ~6,200 veh/day/direction (Oyekanmi et al., 2020); two-way consistent with (World Bank, 2008)
Heavy vehicle share	p_{HGV}	0.159	Fixed	Conservative baseline; Lokoja–Abuja WIM: 30.5% (Oyekanmi et al., 2020); Lagos–Ibadan: 13.7% (World Bank, 2008)
Traffic growth	g	4%/yr	2–6%/yr	Baseline + sensitivity
ESA/HGV (weak enf.)	ESA _{HGV}	10.0	1.6–10.0	TRL proxy
Climate multiplier	k_{clim}	1.15	1.00–1.30	Humid tropical assumption
Initial CI	CI ₀	100	Fixed	Comparative assumption
Initial backlog frac.	—	0.15	Fixed	Conservative; national ≈33% (Foster and Pushak, 2011)
Initial backlog	—	15 km	Fixed	15% of 100 km network
(b) Deterioration rates (CI points/year)				
Condition band	Variable	Baseline	Notes	
Good (CI ≥ 70)	r_{good}	2.2	Calibrated to overlay intervals	
Mid (55 ≤ CI < 70)	r_{mid}	3.0	Calibrated	
Poor (40 ≤ CI < 55)	r_{poor}	4.2	Calibrated	
Very Poor (CI < 40)	r_{vpoor}	6.0	Calibrated	
(c) Maintenance thresholds and recovery				
Parameter	Value		Notes	
Minor repair threshold (CI _{minor})	65		Frozen decision rule	
Overlay threshold (CI _{overlay})	55		Frozen decision rule	
Reconstruction threshold (CI _{recon})	40		Frozen decision rule	
Minor CI recovery (ΔCI _{minor})	+4 pts		Range 3–6	
Overlay CI recovery (ΔCI _{overlay})	+28 pts		Calibrated; cap to 90	
Reconstruction recovery	Reset 100		Full restoration	
Min. interval, minor repairs	3 yr		Fixed	

Table 3.1: Baseline parameter values and sensitivity ranges (*continued*).**(d) Material-specific parameters (SD model coupling values)**

Material	f_{mat}	TTL (yr)	Unit cost (GBP/km)	Unit rehab. CO₂ (t/km)
Conventional asphalt	1.00	5.0	21,257	23.51
Sealed laterite	1.35	3.3	11,821	3.83
Plastic-modified asphalt	0.75	7.0	26,067	24.77

(e) SD model initial conditions and budget

Parameter	Value	Notes
Total network km	100 km	Stylised aggregate
Baseline annual budget	£844,794/yr	Model-anchored; see Section 3.3.2
Baseline material mix	0.70/0.20/0.10	Asphalt/laterite/plastic
ODM (weak enforcement)	1.0	Baseline
ODM (strong enforcement)	0.7	Enforcement scenario

3.2 Asset-Level Digital Twin in MATLAB

3.2.1 Model Purpose and Initial Condition

The asset-level model simulates a stylised road section over 20 years without sensor feeds, parameterised from literature-derived values and calibrated assumptions. The primary state variable is the Condition Index CI_t , bounded 0–100. All three materials are initialised at $CI_0 = 100$ to evaluate pathways from matched starting conditions. Condition updates annually as:

$$CI_{t+1} = \max(0, \min(100, CI_t - \Delta CI_t + R_t)) \quad (1)$$

where ΔCI_t is annual deterioration and R_t is condition recovery from any intervention in year t .

3.2.2 Traffic Loading

Annual ESAs are derived from base traffic level, heavy-vehicle share, and annual growth rate (Table 3.1). The baseline of 13,667 vehicles/day two-way is consistent with the Lokoja–Abuja WIM survey (~6,200 veh/day/direction) (Oyekanmi et al., 2020). The modelled HGV share of 15.9% is a conservative baseline sitting between the Lagos–Ibadan lower bound (13.7%) and the Lokoja–Abuja corridor value (30.5%); a higher share would amplify deterioration and strengthen the enforcement finding. Overloading is incorporated via an ESA-per-HGV factor adjusted by the overload-enforcement multiplier.

3.2.3 Deterioration Model

Annual deterioration is modelled as:

$$\Delta CI_t = r_b \cdot f_{\text{traffic},t} \cdot f_{\text{climate}} \cdot f_{\text{material}} \quad (2)$$

where r_b is the baseline deterioration rate, $f_{\text{traffic},t}$ scales deterioration by annual traffic loading, f_{climate} is a climate multiplier for humid tropical exposure, and f_{material} captures the relative durability of each surfacing material. Traffic scaling follows:

$$f_{\text{traffic},t} = \left(\frac{\text{ESA}_t}{\text{ESA}_{\text{ref}}} \right)^\alpha \quad (3)$$

where α is the loading exponent. The piecewise r_b across four bands captures accelerating decay as condition declines, consistent with empirical pavement-management traditions when mechanistic data are unavailable (Santero et al., 2011a; Harvey et al., 2016). The reference ESA value ESA_{ref} is computed using the non-overloaded ESA/HGV factor (1.6), so that $f_{\text{traffic},t}$ represents the multiplicative damage amplification relative to a normally loaded baseline; at the overloaded baseline this yields $f_{\text{traffic},0} = (10.0/1.6)^{0.3} = 1.73$, rising to 2.17 by year 20 as traffic grows. This is consistent with the stored reference value of 634,531 ESA/year in `config_used.json`.

The material factor f_{material} is a dimensionless calibration parameter that reproduces literature-consistent first-intervention timing for each surface type. Conventional asphalt is the reference ($f_{\text{mat}} = 1.00$). Sealed laterite is more moisture-sensitive and deteriorates faster under tropical conditions, yielding $f_{\text{mat}} = 1.35$, consistent with the shorter 5–9 year reseal cycles reported in Pinard et al. (2003) and Kenya Ministry of Transport, Infrastructure, Housing and Urban Development (2017).

Plastic-modified asphalt has demonstrably improved rutting resistance and high-temperature stability (Xu et al., 2022), calibrated to $f_{\text{mat}} = 0.75$ to reproduce first-overlay intervals of 10–15 years. These values are calibration parameters rather than direct empirical measurements; their sensitivity is examined in Section 3.6.

3.2.4 Maintenance Triggers and Interventions

Three condition-triggered classes are used: minor repair, overlay/reseal, and reconstruction. Each has a CI threshold, recovery magnitude, and associated cost and CO₂. A minimum interval between minor repairs prevents unrealistic clustering. The condition-driven logic allows maintenance timing to emerge endogenously rather than from fixed service-life assumptions, directly addressing the methodological weakness identified by Santero et al. (2011a), who showed that fixed maintenance schedule assumptions dominate pavement LCA sensitivity more than material emission factors.

3.2.5 Cost and CO₂ Accounting

Life-cycle agency cost is discounted as:

$$\text{NPV} = \sum_{t=0}^T \frac{C_t}{(1+d)^t} \quad (4)$$

where d is the discount rate and $T = 20$ years. Construction-phase (embodied) CO₂ is accumulated without discounting:

$$E_{\text{con}} = \sum_{t=0}^T E_t \quad (5)$$

Use-phase rolling-resistance CO₂ at the asset level is computed annually with a condition-index-dependent rolling resistance coefficient. The baseline method follows Trupia et al. (2017) and Chatti and Zaabar (2012), extended with a roughness scaling factor $\varphi(\text{CI}_t)$:

$$E_{\text{use},t} = \text{RRC}_m \cdot \varphi(\text{CI}_t) \cdot F_{\text{mix}} \cdot \text{AADT}_t \cdot \lambda \cdot L \cdot 365 \cdot \text{EF}/1000 \quad (6)$$

$$\varphi(\text{CI}) = 1 + \beta \cdot \max\left(0, \frac{100 - \text{CI}}{15}\right), \quad \beta = 0.014 \quad (7)$$

where $\beta = 0.014$ is the mixed-fleet roughness-fuel sensitivity from Chatti and Zaabar (2012) NCHRP 720 Table 3-5 (LV 0.015, HGV 0.010, weighted 84.1/15.9%). The CI-to-IRI approximation $\text{IRI} = (100 - \text{CI})/15$ m/km anchors to new pavement (CI 100, IRI 0) and the overlay trigger (CI 55, IRI 3.0 m/km), consistent with fair-condition thresholds (Transport Research Laboratory, 1993). A linear form is adopted rather than the log-linear AASHTO relationship, which requires calibrated IRI distributions not available for this context; the linear proxy is fully parameterised by the two model-defined anchor points. Figure 3.1 illustrates the scaling function and the resulting cumulative use-phase CO₂ trajectories, showing how laterite's shorter service cycles sustain a lower mean in-service CI and thereby accumulate greater rolling-resistance emissions than its baseline RRC value alone would predict.

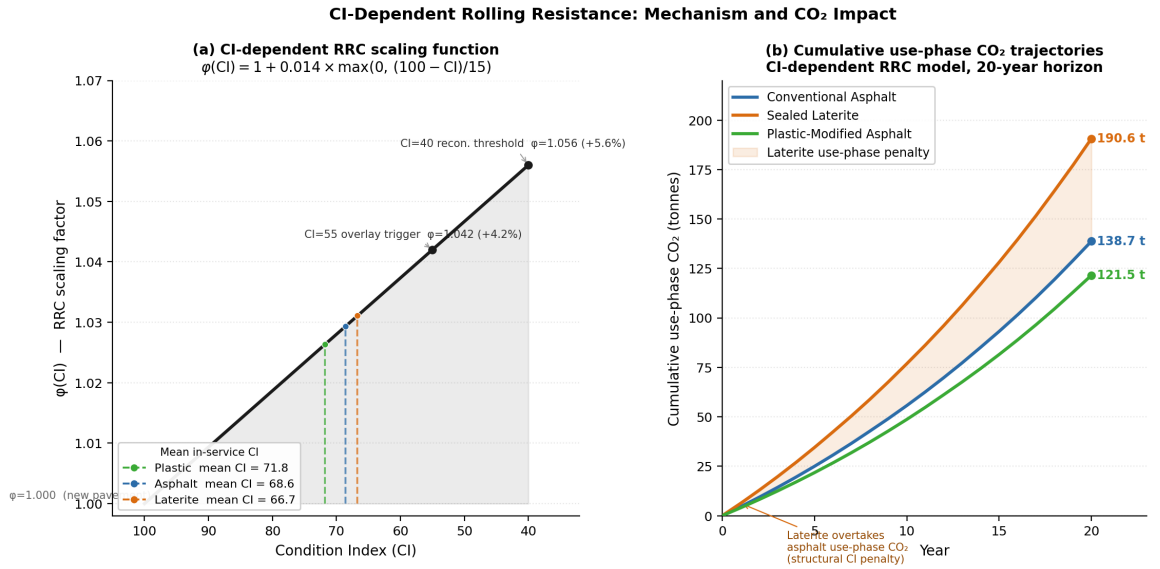


Figure 3.1: CI-dependent RRC mechanism. (a) The $\phi(CI)$ scaling function (Eq. (7)); dashed lines show each material's mean in-service CI. (b) Cumulative use-phase CO₂ trajectories; laterite overtakes asphalt because its shorter TTL sustains a lower mean CI, amplifying rolling-resistance emissions despite having the same baseline RRC.

The SD network model uses fixed baseline RRC values because the Vensim structure tracks aggregate km stocks rather than per-kilometre CI trajectories; the CI-dependent correction applies only to asset-level outputs. Other symbols are as defined in Section 3.3.4; $\lambda = 0.5$, $L = 1$ km.

Whole-life asset-level CO₂ is $E_{wl} = E_{con} + \sum_{t=1}^T E_{use,t}$. Whole-life totals are 335.4 t (asphalt), 253.9 t (plastic), and 236.7 t (laterite). RRC values carry $\pm 25\%$ inter-study uncertainty; applying this range preserves the whole-life hierarchy (laterite < plastic < asphalt) in all cases.

3.2.6 Implementation

The MATLAB model consists of modular functions for traffic, deterioration, maintenance triggering, cost and emissions, NPV, and result export. The digital-twin-enabled framework workflow is shown in Figure 3.2.

Digital Twin Workflow

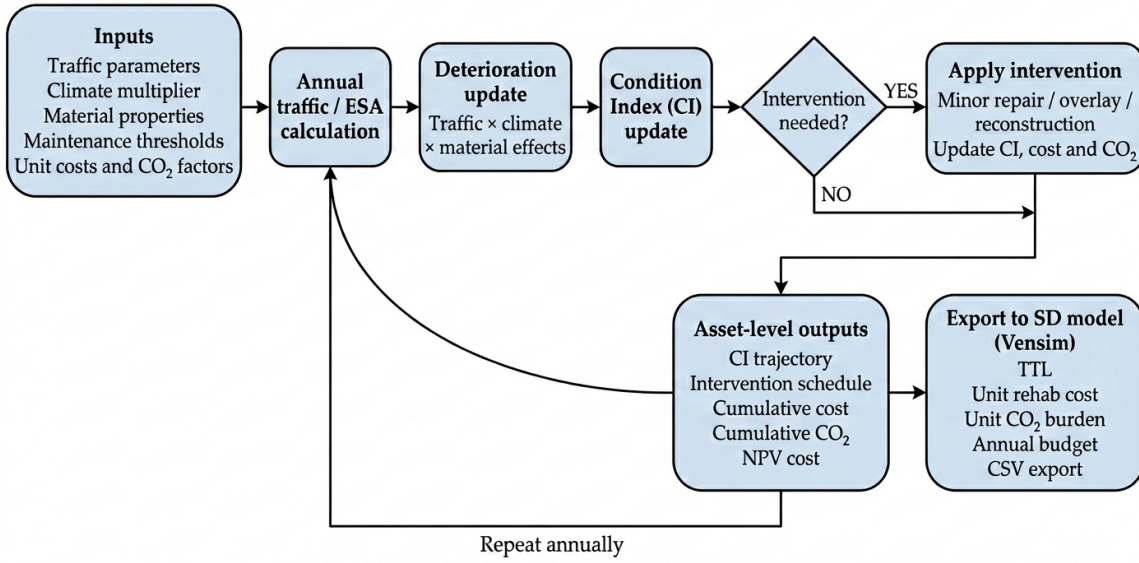


Figure 3.2: Digital-twin-enabled framework workflow: inputs flow through deterioration, condition update, and threshold-triggered intervention to produce asset-level outputs exported to Vensim.

3.3 Network-Level System Dynamics Model in Vensim

3.3.1 Stock-Flow Structure and Initial Conditions

The SD model represents a stylised 100 km network. Main stocks are Acceptable km per material, Backlog km, Cumulative Cost, and Cumulative CO₂. The model is initialised with 15 km backlog (15%) and 85 km acceptable, divided by the baseline material mix. This is conservative relative to the national figure of ~33% (Foster and Pushak, 2011), because the network represents a maintained federal corridor. Sensitivity to this assumption is in Table D.1.

For each material m :

$$\text{Acceptable_km}_m = \text{INTEG}(\text{Rehab}_m - \text{Deterioration}_m, \text{Initial_acceptable_km}_m) \quad (8)$$

$$\text{Backlog_km} = \text{INTEG}(\sum_m \text{Deterioration}_m - \sum_m \text{Rehab}_m, \text{Initial_backlog_km}) \quad (9)$$

The stock-flow structure is shown in Figure 3.3.

3.3.2 Rehabilitation and Budget Logic

Material-specific deterioration flows are:

$$\text{Deterioration}_m = \frac{\text{Acceptable_km}_m}{\text{TTL}_m^{\text{eff}}}, \quad \text{TTL}_m^{\text{eff}} = \frac{\text{TTL}_m}{\text{ODM}} \quad (10)$$

where ODM is the overload-damage multiplier. Annual rehabilitation capacity is budget-constrained:

$$\text{Rehab_capacity_km} = \frac{\text{Annual_Budget}}{\sum_m s_m \cdot c_m} \quad (11)$$

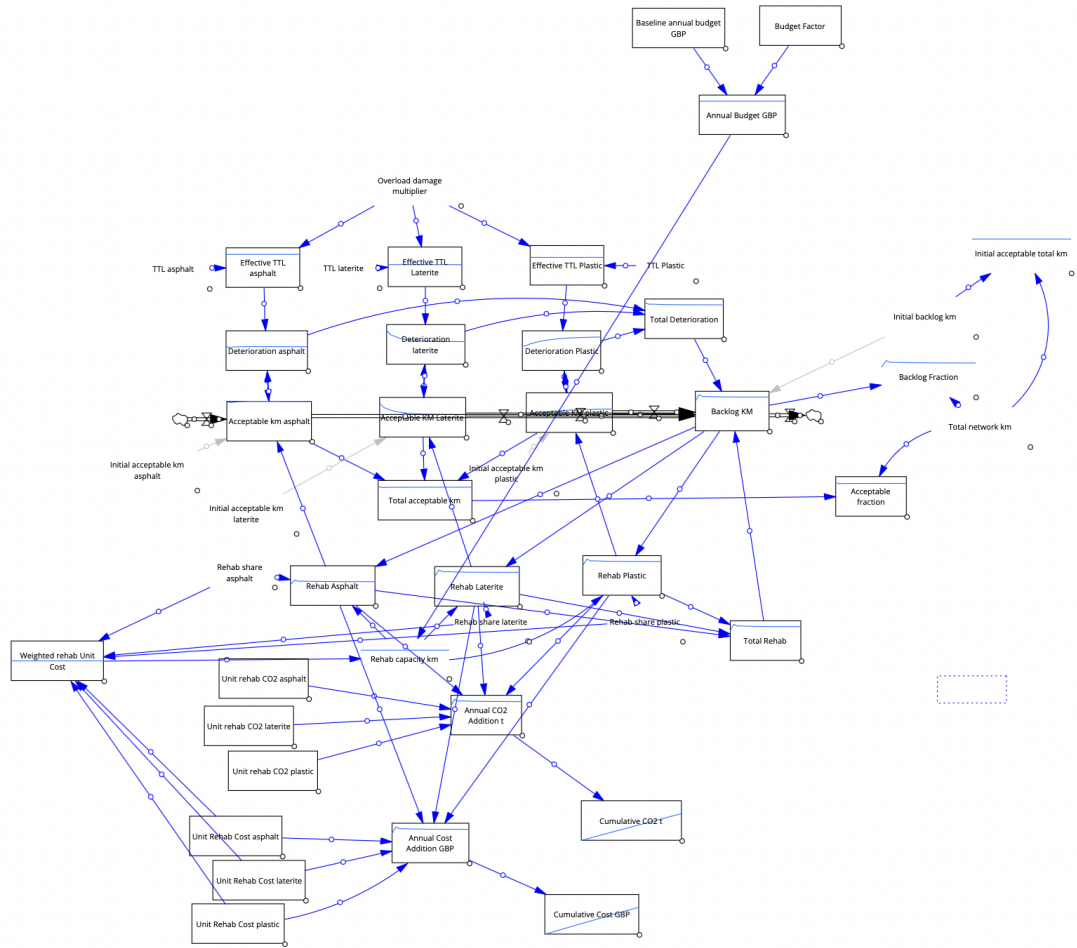


Figure 3.3: System dynamics stock-flow structure: material-specific acceptable-condition stocks deteriorate into backlog; rehabilitation draws from backlog subject to budget constraints; cost and CO₂ accumulate separately.

$$\text{Desired_rehab_rate}_m = \frac{\text{Backlog_km} \cdot s_m}{\tau_{\text{rehab}}}, \quad \tau_{\text{rehab}} = 1 \text{ year} \quad (12)$$

$$\text{Rehab}_m = \min(\text{Desired_rehab_rate}_m, \text{Rehab_capacity_km} \cdot s_m) \quad (13)$$

where s_m is the rehabilitation share, c_m is unit rehabilitation cost for material m , and τ_{rehab} is a first-order rehabilitation response time that converts the backlog stock (km) into a dimensionally consistent flow (km/year). Setting $\tau_{\text{rehab}} = 1$ year ensures unit consistency between the backlog stock and the rehabilitation flow and aligns with the Vensim model's dimensional fix; because the model uses annual time steps, the numerical outputs are identical to the pre-fix formulation. Rehabilitation shares s_m are fixed at the baseline material mix throughout each scenario, so each material receives rehabilitation in proportion to its network share.

This rule is analytically tractable and ensures consistent scenario comparison. Its limitation is that it may under-represent prioritisation by condition severity; in S4 it simultaneously allocates 60% of network share and 60% of rehabilitation capacity to laterite. A priority-based allocation variant is implemented separately (Section 4.3).

To test whether the laterite inversion is an allocation artefact, a priority-based variant directs reha-

bilitation in proportion to each material's current deterioration rate:

$$\pi_m = \frac{\dot{D}_m}{\sum_{m'} \dot{D}_{m'}} = \frac{A_m / \tau_m}{\sum_{m'} A_{m'} / \tau_{m'}} \quad (14)$$

where π_m is the priority weight for material m , A_m is its current acceptable-condition stock (km), and τ_m is its TTL. This logic directs rehabilitation effort to whichever material is deteriorating fastest at each time step, a defensible operational rule consistent with condition-severity-based maintenance management. The priority model uses the same scenario network compositions and budget parameters as the proportional model; the only change is the allocation rule. All seven scenarios are re-run under priority allocation and year-20 backlog outcomes are compared with the proportional baseline in Section 4.2.

The baseline annual budget (£844,794/yr) is derived from the MATLAB annualised cost under the baseline mix. Two structural features make the non-binding result partly a consequence of this anchoring: the budget matches the model's own rehabilitation demand by construction, and dividing this annualised budget by per-event unit costs (£21,257/km) yields a notional capacity (~43 km/yr) that far exceeds annual deterioration demand (~5–6 km/yr). See Section 5.2.

The SD model represents only reactive rehabilitation of roads that have entered backlog; it does not include a preventive maintenance pathway. Preventive maintenance is omitted not because it is unimportant, but because the model is designed to reflect the reactive maintenance environment documented for the study context (Foster and Pushak, 2011; Yakubu et al., 2023). Including a preventive pathway would require additional calibration data not currently available and is identified as a priority for future work.

3.3.3 Coupling with the Digital Twin

The SD model is parameterised from MATLAB outputs: TTL values, unit rehabilitation costs, unit CO₂ burdens, and baseline budget. The coupling is one-way: the SD model inherits material behaviour from the asset model but does not update it during simulation. Asset-level condition-band logic is compressed into a single TTL scalar per material, a unit cost, and a unit CO₂ value. The laterite TTL (3.33 yr) is grounded in low-volume road literature; the TTL sensitivity analysis (Table 4.8) quantifies how conclusions shift across the plausible range. Coupling variables are documented in Table 3.2.

Table 3.2: MATLAB $\rightarrow$ Vensim one-way coupling mapping.

MATLAB output	Definition	Vensim variable	Role in SD model	Notes
mean_overlay_ interval_year	Average years between overlays/reseals	TTL_asphalt, TTL_laterite, TTL_plastic	Controls deterioration flow	Time-to-backlog proxy
cost_per_km_ year_gbp	Annual agency cost intensity (GBP/km)	Unit rehab cost	Defines rehabilitation capacity	Combined with adoption shares
co2_per_km_ year_t	Annual construction-phase CO ₂ intensity (t/km)	Unit CO ₂	Annual embodied CO ₂ from rehab	Asset to network CO ₂
npv_cost_gbp	20-yr discounted cost	Reference metric	Plausibility check	Not a flow driver
total_co2_t	20-yr construction-phase CO ₂ (t)	Reference metric	Reporting	Not a flow driver
n_minor, n_overlay	Intervention counts	Policy interpretation	Backlog dynamics	Discussion support

3.3.4 Cost and Emissions Accumulation

Annual construction-phase cost and CO₂ flows are computed as $\sum_m \text{Rehab}_m \cdot c_m$ and $\sum_m \text{Rehab}_m \cdot e_m$ respectively, and accumulated as INTEG stocks initialised at zero.

Use-phase rolling-resistance CO₂

A second CO₂ stock accumulates use-phase rolling-resistance emissions from vehicle fuel consumption on the acceptable-condition network. The method follows Trupia et al. (2017) and Chatti and Zaabar (2012):

$$\dot{E}_{\text{use}} = \sum_m A_m \cdot \text{RRC}_m \cdot F_{\text{mix}} \cdot \text{VKT}_{\text{yr}} \cdot \text{EF} \quad (15)$$

where A_m is acceptable-condition stock (km), RRC_m is the rolling resistance coefficient, F_{mix} is the fleet-average fuel factor (L/veh-km per unit RRC), VKT_{yr} is annual vehicle-km per km of road, and $\text{EF} = 2.64 \text{ kg CO}_2/\text{L}$ (IPCC, 2006). RRC values are 0.0120 (asphalt), 0.0105 (plastic), 0.0165 (laterite) (Trupia et al., 2017; Chatti and Zaabar, 2012). The fleet fuel factor is $F_{\text{mix}} = (1 - p_{\text{HGV}}) \times 0.040 + p_{\text{HGV}} \times 0.150 = 0.0437 \text{ L/veh-km/unit-RRC}$ (Chatti and Zaabar, 2012). VKT grows at 4%/year. RRC values are fixed at good-condition baselines in the network model; the residual limitation is discussed in Section 5.4. Roads in backlog are excluded from use-phase accumulation. Scenarios with less backlog (S6, S5) therefore generate *more* use-phase CO₂ than high-backlog scenarios (S4), because backlog roads in reality carry traffic at elevated roughness that the model does not capture; the backlog-exclusion assumption makes the CO₂ advantage of enforcement-heavy scenarios appear smaller than it would under a full-inclusion model. The ranking is conservative rather than neutral. The two CO₂ stocks, construction-phase and use-phase, are summed to yield whole-life CO₂. Primary outputs are backlog km, acceptable fraction, cumulative agency cost, cumulative construction-phase CO₂, cumulative use-phase CO₂, and cumulative whole-life CO₂.

3.4 Scenario Design

Scenarios vary three policy levers: budget level (factors 0.7, 1.0, 1.3), material adoption mix (baseline 0.70/0.20/0.10 asphalt/laterite/plastic; laterite-heavy S4 at 60%; plastic-heavy S6 at 60%), and overload enforcement (binary: off/on). The overload damage multiplier of 0.7 represents a moderate enforcement improvement: under the fourth-power law (Transport Research Laboratory, 1993), if half of overloaded vehicles are brought into legal compliance, network-average damage falls by approximately 25–35%.

Seven named scenarios are defined in Table 3.3. S1–S2 isolate budget variation; S3–S4 test alternative material mixes; S5 tests enforcement alone; S6 combines enforcement with plastic-push to test lever interaction, addressing RQ3. A time-to-crisis metric records the year at which backlog fraction first exceeds 20%.

In addition to the seven named scenarios, two structural robustness analyses are conducted under the baseline material mix (0.70/0.20/0.10) and weak enforcement (ODM = 1.0): (i) an extended budget-factor sweep over factors 0.10, 0.20, 0.30, 0.40, 0.41, 0.50, 0.70, 1.00, and 1.30, designed to identify the budget factor at which the rehabilitation constraint first becomes binding; and (ii) an initial backlog sensitivity run at fractions of 5%, 15%, 25%, and 35% of the 100 km network. For the budget sweep, a budget factor is defined as *binding* when the year-20 backlog diverges from the S0 (factor 1.0) result by more than 0.01 km; this threshold is pre-specified and corresponds to a deviation of approximately 0.06% of total network length, well above numerical precision but small enough to capture genuine behavioural change. Both analyses use S0 parameters except the varied dimension.

A network-level Monte Carlo is also pre-specified, varying two parameters jointly across $N = 1,000$ Latin Hypercube samples (seed 42): ODM (uniform 0.6–1.4) and initial backlog fraction (uniform 0.05–0.35). Scenarios S0, S5, and S6 are evaluated at each sample. This analysis is separate from the asset-level Monte Carlo and tests whether the scenario ordering is robust across the plausible range of enforcement conditions and initial network states. Results are reported in Section 4.4.

Table 3.3: Scenario matrix.

Scenario	Budget factor	Mix (A/L/P)	ODM	Primary purpose
S0 Baseline	1.0	0.70 / 0.20 / 0.10	1.0	Reference
S1 Low-budget	0.7	0.70 / 0.20 / 0.10	1.0	Budget stress
S2 High-budget	1.3	0.70 / 0.20 / 0.10	1.0	Budget increase
S3 Plastic-push	1.0	0.25 / 0.15 / 0.60	1.0	Material substitution
S4 Laterite-heavy	1.0	0.30 / 0.60 / 0.10	1.0	Laterite dominant
S5 Strong enf.	1.0	0.70 / 0.20 / 0.10	0.7	Enforcement only
S6 Combined policy	1.0	0.25 / 0.15 / 0.60	0.7	Enforcement + materials

3.5 Multi-Criteria Decision Analysis

A weighted-sum MCDA combines three lower-is-better criteria: cumulative agency cost, cumulative CO₂, and final backlog fraction as a robustness check on the scenario ranking. Each criterion is min-max normalised as $\tilde{x}_{ik} = (x_k^{\max} - x_{ik}) / (x_k^{\max} - x_k^{\min})$, and the composite score is:

$$S_i = w_1 \tilde{x}_{i,\text{cost}} + w_2 \tilde{x}_{i,\text{CO}_2} + w_3 \tilde{x}_{i,\text{backlog}}, \quad w_1 + w_2 + w_3 = 1, \quad w_k \geq 0 \quad (16)$$

3.5.1 Weight Sampling and Robustness

Cost weight $w_1 \in [0.3, 0.5]$ and emissions weight $w_2 \in [0.2, 0.4]$ are drawn uniformly; $w_3 = 1 - w_1 - w_2$ is accepted if $w_3 \in [0.2, 0.4]$, yielding 10,000 valid samples. The emissions criterion uses construction-phase CO₂ for consistency with the asset-level Monte Carlo. A coherence check using whole-life CO₂ confirmed S6 ranks first in 100% of samples under both formulations.

3.5.2 Budget-Equivalent Framing of Enforcement Benefits

A budget-equivalent analysis translates enforcement-driven backlog reductions into monetary equivalents: Budget Equivalent_{annual} = $\Delta\text{Backlog} \times \bar{c}/T$, where $\Delta\text{Backlog}$ is the year-20 backlog reduction versus S0, $\bar{c}$ is the weighted baseline unit rehabilitation cost, and $T = 20$ years. This framing is valid specifically because the system is deterioration-driven and additional budget produces no backlog improvement within the tested range; in a budget-constrained system the same calculation would not apply. The figures are maintenance-budget analogues of enforcement benefit, not willingness-to-pay in the welfare-economic sense.

3.6 Model Verification, Plausibility Checks, and Sensitivity Analysis

Table 3.4 summarises the principal validation checks, following Barlas (1996) and Sterman (2000).

Table 3.4: Summary of principal validation and plausibility checks.

Check type	Benchmark or expectation	Model result	Verdict
Dimensional consistency	Equations must preserve units across stocks and flows	MATLAB and Vensim equations checked variable-by-variable; rehabilitation flows corrected to divide backlog stock by a 1-year response time	No inconsistencies found
First intervention timing	Asphalt: 8–12 yr; laterite: 5–9 yr; plastic: 10–15 yr	Asphalt yr 10; laterite yr 8; plastic yr 12	All within target windows; laterite sits at the midpoint of the 5–9 yr window (Pinard et al., 2003; Kenya Ministry of Transport, Infrastructure, Housing and Urban Development, 2017)
Corridor condition plausibility	PSR = 3.45 on Lokoja–Abuja corridor (Oyekanmi et al., 2020)	CI ≈ 69 derived from PSR–CI mapping (Eq. (17)); modelled asphalt CI in 65–72 range at 7–10 yr pre-overlay	PSR-derived CI falls within predicted mid-service range; see Section 3.6
Extreme-condition behaviour	Short TTLs should increase backlog rapidly	Monotonic backlog response confirmed over wide TTL range	Directionally credible
Cross-model consistency	S0 SD baseline should reproduce MATLAB-anchored parameters	Year-20 backlog under S0 SD (17.054 km) consistent with steady-state implied by MATLAB-derived parameters; outputs self-consistent within numerical precision	Coupling verified
Budget non-binding test	Capacity check: budget supports ~ 43 km/yr; inflow is 5–6 km/yr	Extended sweep confirms non-binding above budget factor ≈ 0.41 (£346,000/yr); S0/S1/S2 identical at all tested levels	Binding threshold identified; non-binding result robust across budget factors 0.41–1.30

A Monte Carlo simulation ($N = 2,000$, seed 42) samples seven uncertain parameters simultaneously: k_{clim} , g , d , α , unit cost factors, $f_{\text{mat,laterite}}$, and $f_{\text{mat,plastic}}$. Ranges are $\pm 20\%$ of calibrated values (Pinard et al., 2003; Kenya Ministry of Transport, Infrastructure, Housing and Urban Development, 2017).

A variance-based Sobol sensitivity analysis decomposes output variance into direct (S_1) and interac-

tive (S_T) contributions from each parameter. Where $S_T \gg S_1$, influence operates primarily through interactions, a finding invisible to OAT methods. The analysis uses the Saltelli sampling scheme (Saltelli et al., 2010) with $N = 1,024$ base samples (9,216 total evaluations) and the Jansen estimator. It is implemented in Python using a faithful reconstruction of the MATLAB model; cross-verification confirmed agreement within 0.3% on NPV and 0.5% on CO₂. Results are in Section 4.1.6. Important structural parameters (ODM, initial backlog fraction, and rehabilitation shares) are not varied at the asset level; these are addressed in the network-level Monte Carlo (Section 4.4).

Corridor condition plausibility check

Oyekanmi et al. (2020) report a Present Serviceability Rating of PSR = 3.45/5.0 on the Lokoja–Abuja corridor in April 2019. Converting PSR to CI uses a linear correspondence anchored to ORN 31 condition classifications (Transport Research Laboratory, 1993): new pavement (PSR = 5.0, CI = 100) and reconstruction boundary (PSR = 2.0, CI = 40):

$$CI \approx 40 + (PSR - 2.0) \times 20 \quad (17)$$

Applied to PSR = 3.45, Eq. (17) gives $CI \approx 69$. The model predicts asphalt CI in the range 65–72 during the 7–10 years before first overlay is triggered; a CI of 69 corresponds to approximately 8 years into a service cycle, consistent with a well-maintained federal corridor mid-way through its maintenance interval. This converts the previously qualitative “broadly consistent” verdict into a quantitative anchor: the observed PSR maps to a specific position within the modelled trajectory rather than merely within a broad fair-condition band. The equation is approximate: the PSR–CI correspondence is calibrated to AASHTO serviceability thresholds and has not been empirically validated against Nigerian pavement types, but it provides a falsifiable external check absent from simulation-only validation.

Federal network roughness as additional calibration evidence

Three independent Nigerian IRI datasets provide external anchors for the model’s condition scale. Efiom et al. (2021) measured roughness on four federal routes connecting Abuja (FCT) to neighbouring state capitals using calibrated smartphone applications, recording average IRI values of 3.70 m/km (Roadroid) and 4.48 m/km (RoadLab Pro). Popoola et al. (2020) used Federal Ministry of Works Pavement Evaluation Unit data from south-west Nigerian federal roads (2012–2015), finding IRI values of 3.5–4.4 m/km on maintained but deteriorating sections. Road Sector Development Team (2014) calibrated HDM-4 to Nigerian federal road conditions, reporting initial roughness values consistent with this range.

Converting to the model’s CI scale using Eq. (17): IRI 3.70 m/km corresponds to $CI \approx 44.5$ and IRI 4.48 m/km to $CI \approx 33$, near or below the overlay trigger (CI 55) and reconstruction threshold (CI 40). Together with the Lokoja–Abuja PSR anchor ($CI \approx 69$), these datasets bracket the model’s CI operating range from both ends: a well-maintained corridor at the upper boundary, maintenance-deficit corridors at the lower, converting the single-point PSR check into a range-spanning calibration.

3.7 Limitations

Deterioration parameters are calibrated to literature targets, supplemented by Nigerian roughness anchors (Sections 3.6 and 3.6). Traffic follows the Lokoja–Abuja corridor (Oyekanmi et al., 2020); the modelled HGV share (15.9%) is conservative within the 13.7–30.5% inter-corridor range. Environmental scope is CO₂ only; economic scope is agency costs only. The laterite inversion is robust under both proportional and priority-based allocation, but neither incorporates spatial prioritisation or institutional capacity constraints.

4 Results

4.1 Asset-Level Material Comparison

4.1.1 Intervention Frequency and Calibration Validation

Table 4.1 presents the baseline intervention frequency for each material over the 20-year horizon. No reconstruction events occurred for any material, indicating that the modelled maintenance policy prevented the condition index from crossing the reconstruction threshold.

Table 4.1: Intervention frequency and life-cycle outcomes: baseline simulation.

Material	Minor	Overlays	Recon.	1st overlay (yr)	NPV cost (£)	Con. CO ₂ (t)	W-L CO ₂ (t)
Conventional asphalt	3	3	0	Year 10	99,872	195.9	335.4
Sealed laterite	4	4	0	Year 8	79,688	44.7	236.7
Plastic-modified asphalt	2	2	0	Year 12	110,692	132.1	253.9

All three materials satisfy their calibration targets: asphalt first overlay year 10 (target 8–12 yr), laterite year 8 (target 5–9 yr), plastic year 12 (target 10–15 yr), confirming physically plausible maintenance cycles.

4.1.2 Condition Index Trajectories

Figure 4.1 shows the 20-year CI trajectories for all three materials. All three materials remain above the reconstruction threshold (CI = 40) throughout. Plastic-modified asphalt sustains the highest average condition, consistent with its lower deterioration factor. Laterite shows the most active profile, reaching a pre-reseal CI of 55.7 at year 8 compared to 65.3 for both asphalt and plastic at their respective overlay trigger points.

4.1.3 Life-Cycle Cost and CO₂

The 20-year NPV costs are: laterite £79,688, asphalt £99,872, and plastic £110,692. Laterite is 20.2% cheaper than asphalt and 28.0% cheaper than plastic, driven by its substantially lower unit rehabilitation cost (£11,821/km vs £21,257/km for asphalt overlay; see Table 3.2). Plastic carries a 10.8% NPV cost premium over asphalt, reflecting its 20% unit cost uplift despite requiring fewer interventions.

Construction-phase CO₂ is laterite 44.7 t, plastic 132.1 t, asphalt 195.9 t. Including use-phase rolling resistance (Eq. (7)), whole-life totals are laterite 236.7 t, plastic 253.9 t, asphalt 335.4 t; laterite's advantage narrows from 151 t to 99 t. The CI-dependent model reveals a structural penalty: laterite's shorter reseal cycles sustain lower average condition (mean CI 66.7 vs 68.6 for asphalt), accumulating more use-phase CO₂ than a fixed-RRC assumption would predict. This extends the Chatti and Zaabar (2012) framework by identifying service-life-driven CI depression as the mechanism. Plastic

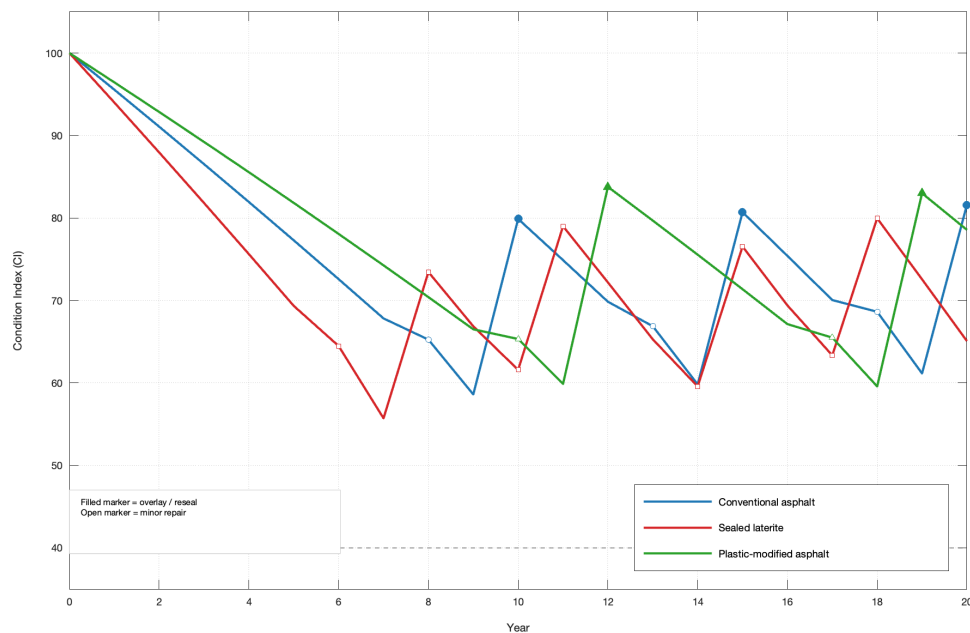


Figure 4.1: Condition index trajectories under baseline traffic and climate conditions (AADT = 13,667, $k_{\text{climate}} = 1.15$, ESA/HGV = 10.0).

retains an 81.5 t whole-life advantage over asphalt because its longer TTL sustains higher condition (mean CI 71.8). Figure 4.2 shows asphalt's 58% construction-phase dominance, contradicting the use-phase-dominated pattern of Santero et al. (2011a).

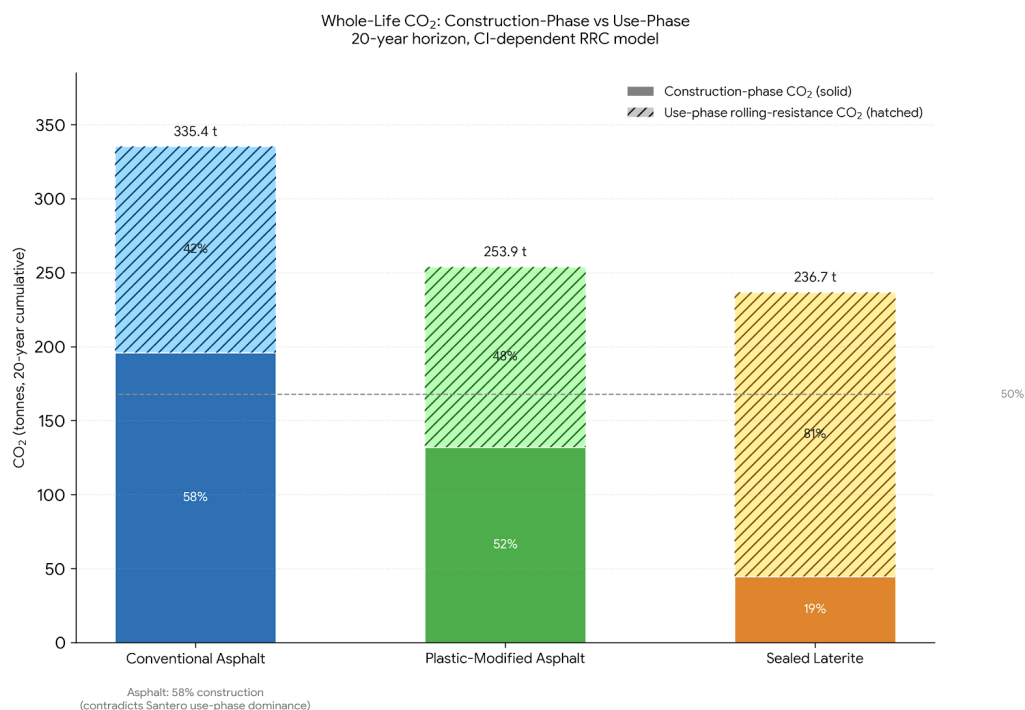


Figure 4.2: Whole-life CO₂ by phase. Solid bars: construction-phase; hatched bars: use-phase rolling-resistance (CI-dependent RRC model). Percentages show each phase's share of the whole-life total. Asphalt's 58% construction-phase share contradicts the use-phase-dominated pattern reported by Santero et al. (2011a) for temperate-climate studies.

For CO₂, plastic-modified asphalt is lower-emitting from year 0, with an immediate 20.7 t advantage

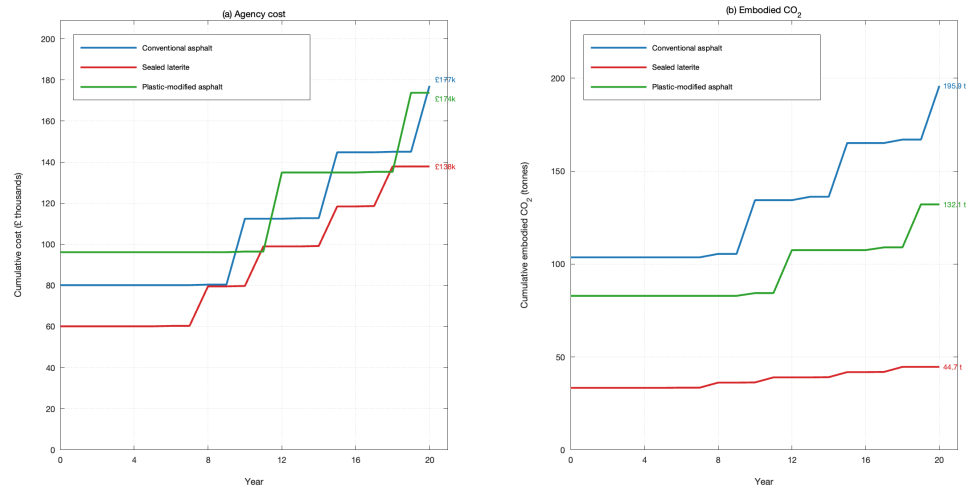


Figure 4.3: Cumulative undiscounted agency cost (left) and embodied CO₂ (right) for all three materials over the 20-year horizon.

growing to 63.8 t by year 20. For cost, plastic begins with a higher construction outlay (£96,203 vs £80,169) but reaches cost parity at year 10 when asphalt’s second overlay triggers. By year 20, plastic is marginally cheaper in undiscounted terms but remains more expensive on an NPV basis (£110,692 vs £99,872); the cost case therefore depends on analysis horizon and discounting convention, examined in Section 4.1.4.

4.1.4 25-Year Horizon Sensitivity

Extending the analysis horizon to 25 years tests whether plastic’s NPV premium persists at longer planning horizons. Over years 21–25, asphalt requires one further overlay at year 25 (adding approximately £1,903 discounted at 12%) while plastic requires none; incremental discounted maintenance is £1,903 for asphalt versus approximately £20 for plastic in this period.

Table 4.2: Horizon sensitivity: 20-year vs 25-year outcomes (construction-phase CO₂ and NPV cost).

Material	NPV yr 20 (£)	NPV yr 25 (£)	Con. CO ₂ yr 20 (t)	Con. CO ₂ yr 25 (t)
Conventional asphalt	99,872	101,775	195.9	261.2
Sealed laterite	79,688	82,411	44.7	67.1
Plastic-modified asphalt	110,692	110,712	132.1	132.1

The 25-year extension produces two clear findings. First, plastic’s undiscounted cost advantage over asphalt grows from £3,400 at year 20 to £37,000 at year 25, as asphalt requires a fourth overlay (£32,075 undiscounted) while plastic requires none; the cumulative cost advantage compounds decisively on longer horizons. Second, plastic’s CO₂ advantage nearly doubles from 63.8 t to 129 t as asphalt’s fourth overlay adds 65.3 t of construction-phase emissions. On a discounted-NPV basis the ranking is unchanged at 12%, because a future avoided overlay is worth only £1,900 in present-value terms; the NPV case for plastic therefore rests on the discount rate assumption, while the undiscounted and CO₂ cases strengthen unambiguously with horizon.

4.1.5 Monte Carlo Uncertainty Analysis

Table 4.3 presents the Monte Carlo results ($N = 2,000$). Laterite’s 95th-percentile CO₂ (88.9 t) falls below the 5th percentile of plastic (109.0 t), confirming a robust advantage (Santero et al., 2011a;

Pinard et al., 2003). Laterite retains the lowest median NPV (£89,874), though its upper tail overlaps with other materials; its cost advantage is strong but not absolute.

Table 4.3: Monte Carlo summary statistics ($N = 2,000$, seed = 42).

Metric	Material	Median	5th pct	95th pct
NPV Cost (GBP)	Conventional asphalt	108,754	74,950	158,555
	Sealed laterite	89,874	61,044	135,945
	Plastic-modified asphalt	117,979	83,756	165,796
Construction-phase CO ₂ (t)	Conventional asphalt	197.7	165.1	311.5
	Sealed laterite	47.6	41.8	88.9
	Plastic-modified asphalt	133.6	109.0	204.5
First overlay (yr)	Conventional asphalt	9.0	6.0	12.0
	Sealed laterite	7.0	4.0	9.0
	Plastic-modified asphalt	10.0	7.0	15.0

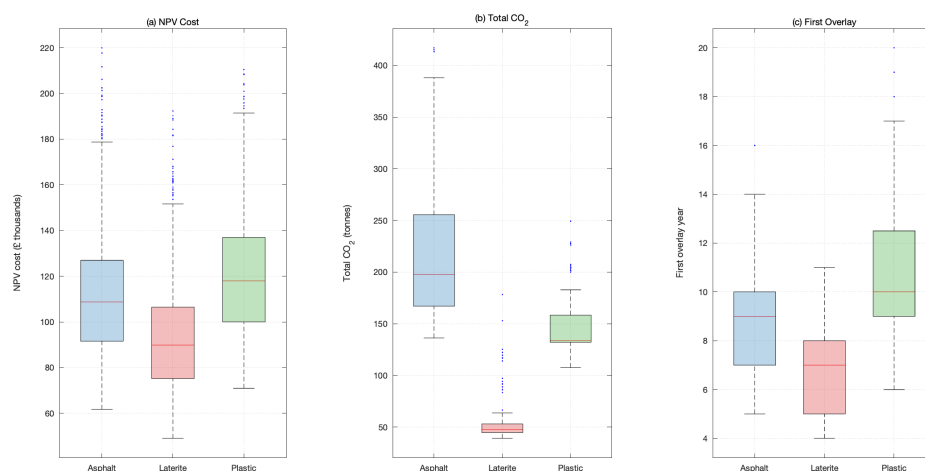


Figure 4.4: Monte Carlo uncertainty distributions ($N = 2,000$) for NPV cost, construction-phase CO₂, and first overlay year across six sampled parameters (ODM and initial backlog varied separately in the network-level Monte Carlo, Table 4.10).

4.1.6 Sobol Variance-Based Sensitivity Analysis

Table 4.4 reports first-order (S_1) and total-order (S_T) Sobol indices for NPV cost and construction-phase CO₂. The OAT tornado chart (Figure 4.5) identified cost level and α as the two most influential parameters. Sobol analysis confirms this for NPV but reveals a fundamentally different picture for CO₂.

Table 4.4: Sobol sensitivity indices. S_1 = first-order; S_T = total-order (includes interactions). $N = 1,024$ base samples; Jansen estimator (Saltelli et al., 2010). Values below 0.01 shown as —. High $S_T - S_1$ = primarily interactive.

Output	k_{clim}		α		Cost factor		$f_{\text{mat}}^\dagger$	
	S_1	S_T	S_1	S_T	S_1	S_T	S_1	S_T
Asphalt NPV	—	0.06	0.03	0.06	0.24	0.81	—	—
Laterite NPV	—	0.06	0.04	0.06	0.19	0.65	0.38	0.14
Plastic NPV	—	0.04	0.02	0.04	0.27	0.83	0.03	0.07
Asphalt CO ₂	0.06	0.62	0.37	0.61	—	—	—	—
Laterite CO ₂	—	0.28	0.11	0.31	—	—	0.84	0.54
Plastic CO ₂	—	0.40	0.20	0.41	—	—	0.14	0.65

$^\dagger f_{\text{mat}}$: laterite factor for laterite outputs; plastic factor for plastic outputs.

NPV. Cost factor dominates NPV for all materials ($S_T = 0.65$ – 0.83), consistent with the OAT result. The large interaction component ($S_T - S_1 \approx 0.46$ – 0.57) indicates that cost uncertainty amplifies the effects of other parameters rather than acting independently. For laterite NPV specifically, $f_{\text{mat, laterite}}$ has the largest direct effect ($S_1 = 0.38$), confirming that intervention timing is the primary material-specific cost driver.

CO₂: the key Sobol finding. The climate multiplier has $S_1 = 0.06$ but $S_T = 0.62$ for asphalt CO₂: 91% of its total influence operates through interactions with α . The physical mechanism is multiplicative: in high tropical deterioration conditions, a higher loading exponent accelerates CI decline nonlinearly, producing more frequent overlays and compounding construction-phase CO₂. OAT would attribute CO₂ variance primarily to α alone and leave k_{clim} 's role almost invisible. The same pattern holds for plastic CO₂ (k_{clim} : $S_1 = 0.00$, $S_T = 0.40$). For laterite CO₂, $f_{\text{mat, laterite}}$ dominates directly ($S_1 = 0.84$), because the number of chip-seal reseals, governed primarily by the material factor in the laterite case, determines embodied carbon.

The contrast between laterite and the bituminous materials is structurally informative. For asphalt and plastic, overlays occur only two or three times over 20 years; whether a second or third overlay falls within the horizon depends on the compounded interaction of climate, traffic growth, and loading exponent. For laterite, reseals occur four or five times; the frequency is so directly controlled by f_{mat} that climate-traffic interaction effects become secondary relative to the dominant TTL driver. In other words, the parameter structure of CO₂ uncertainty differs by material not because the physics differs, but because the maintenance frequency places each material in a different sensitivity regime: high-frequency maintenance (laterite) is variance-dominated by the material factor; low-frequency maintenance (asphalt, plastic) is variance-dominated by the compounded interaction of climate and loading. This has an implication for how LCA uncertainty should be reported: pooling all materials into a single sensitivity framework obscures material-specific variance structures that change the ranking of which parameters matter most.

The full decomposition across all seven parameters is shown in Figure 4.6.

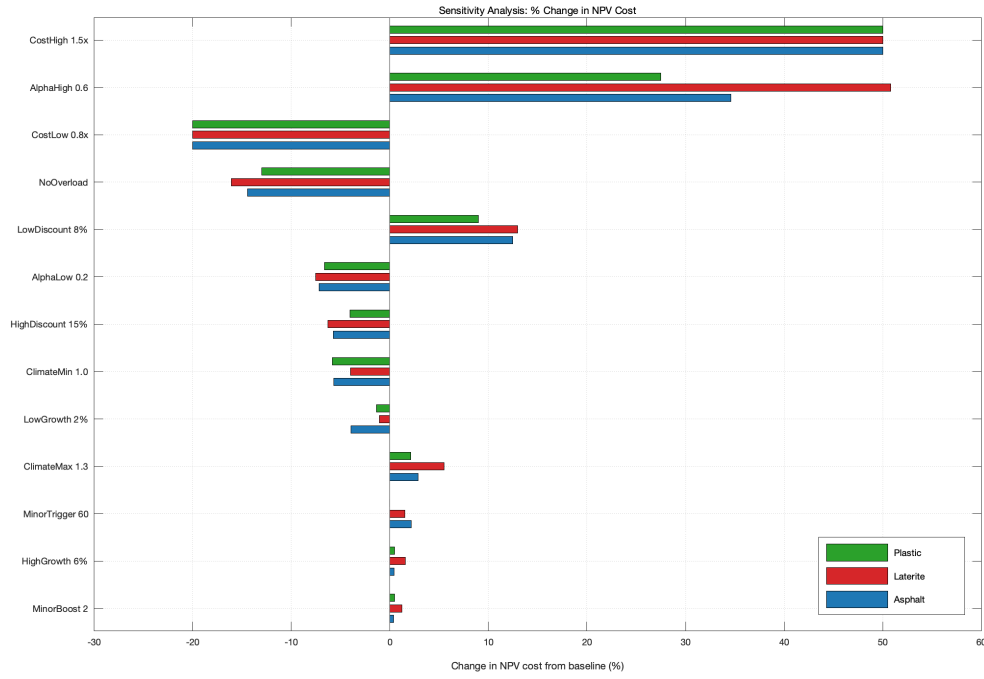


Figure 4.5: One-at-a-time sensitivity analysis: percentage change in NPV cost from baseline. Compare with Table 4.4: OAT correctly identifies the dominant NPV drivers but understates climate multiplier's role in CO₂ uncertainty, which operates almost entirely through interaction with α .

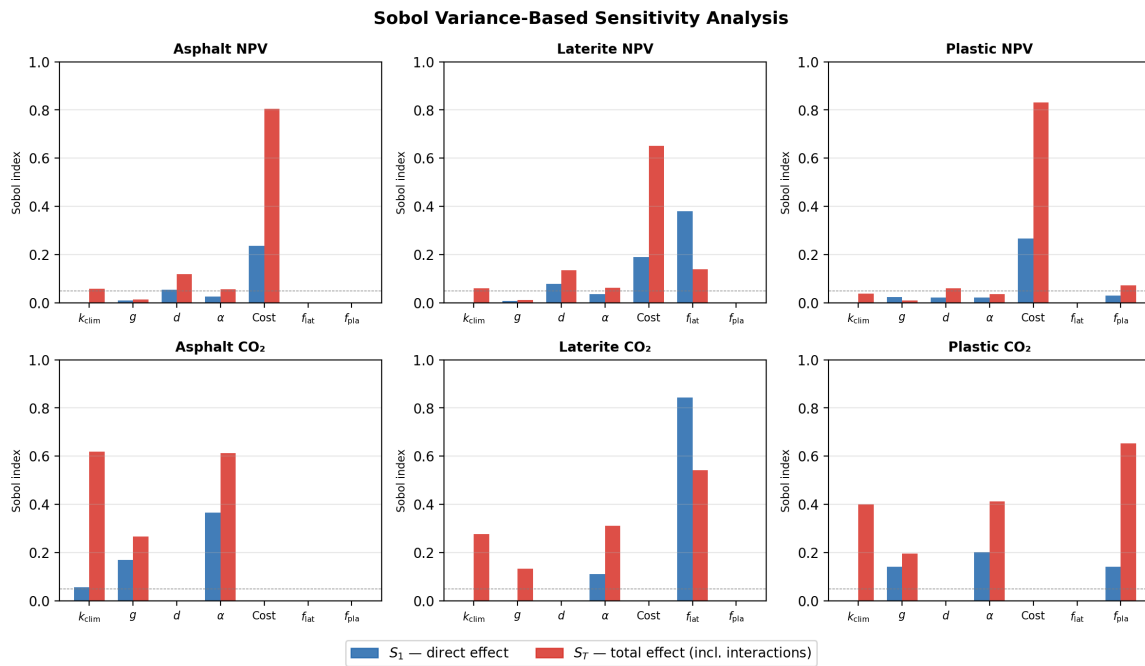


Figure 4.6: Sobol sensitivity analysis: first-order (S_1 , blue) and total-order (S_T , red) indices for NPV and whole-life CO₂ for each material. Parameters with $S_T \gg S_1$ act primarily through interactions. The k_{clim} – α interaction dominates CO₂ uncertainty for asphalt and plastic; $f_{mat, laterite}$ drives laterite CO₂ directly.

4.2 Network-Level Scenario Analysis

4.2.1 Backlog Trajectories and the Non-Binding Budget Result

Figure 4.7 shows the 20-year network backlog trajectories. Table 4.5 summarises outcomes at years 1, 10, and 20.

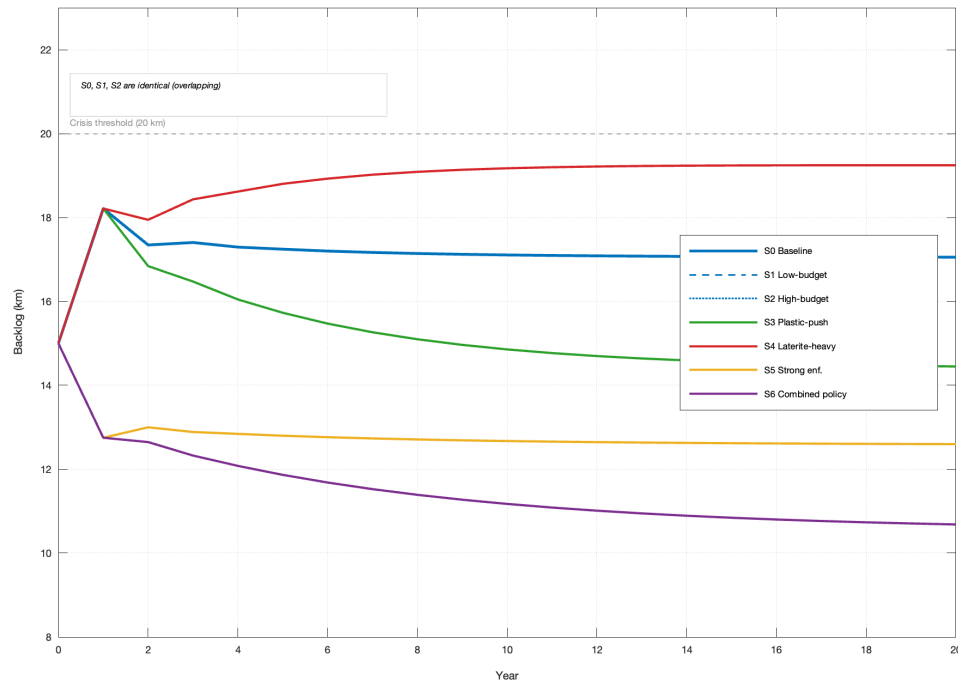


Figure 4.7: Network backlog trajectories for all seven scenarios. S0, S1, and S2 overlap completely, confirming the budget is non-binding; the binding constraint is deterioration inflow.

Table 4.5: Network-level scenario outcomes. Con. CO₂ = construction-phase; W-L CO₂ = whole-life.

Scenario	Bklg yr1	Bklg yr10	Bklg yr20	Cum. cost (£M)	Con. CO ₂ (t)	Use CO ₂ (t)	W-L CO ₂ (t)
S0 Baseline	18.21	17.11	17.05	6.79	6,734	11,194	17,929
S1 Low-budget	18.21	17.11	17.05	6.79	6,734	11,194	17,929
S2 High-budget	18.21	17.11	17.05	6.79	6,734	11,194	17,929
S3 Plastic-push	18.21	14.86	14.45	6.94	6,504	10,674	17,179
S4 Laterite-heavy	18.21	19.17	19.24	6.04	4,440	12,067	16,507
S5 Strong enf.	12.75	12.67	12.60	5.09	5,052	11,804	16,855
S6 Combined policy	12.75	11.17	10.68	5.26	4,934	11,248	16,182

S0, S1, and S2 are identical at every time step: rehabilitation capacity (~ 43 km/yr) structurally exceeds deterioration inflow ($\sim 5\text{--}6$ km/yr), so budget variation of $\pm 30\%$ produces no change. The structural implication for scenario comparison is important: when budget is non-binding, the only variable driving backlog differences across scenarios is the TTL-weighted deterioration flow. S4's higher backlog is therefore a pure TTL effect — laterite's 3.33 yr TTL generates a deterioration flow of approximately 5.1 km/yr per 17 km of acceptable-condition laterite stock, which at 60% network share exceeds the deterioration rates of all asphalt-mix scenarios. No budget level within the non-binding region can compensate for a faster deterioration inflow; only a longer TTL or a lower laterite network share can. The implications are in Section 5.2.

To establish the precise boundary of this result, the model was run across a wider range of budget factors. Table 4.6 reports year-20 backlog and rehabilitation capacity at budget factors from 0.10 to

1.30.

Table 4.6: Budget threshold analysis (selected factors).

Budget factor	Annual budget (£)	Rehab capacity (km/yr)	Yr-20 backlog (km)	Budget binding?
0.30	253,438	12.8	37.7	Yes
0.40	337,918	17.0	17.3	Yes (marginally)
0.41	346,366	17.4	17.05	No (threshold)
0.70	591,356	29.8	17.05	No
1.00	844,794	42.6	17.05	No

The non-binding result holds at all budget factors above approximately 0.41 (£346,000/yr). Below this, rehabilitation capacity cannot clear backlog demand. The tested range 0.7–1.3 sits well above this threshold, explaining the identity of S0, S1, and S2.

Alternative Budget Anchor Sensitivity

A demand-based anchor (£361,600/yr) confirms the result is anchor-specific: backlog rises to 37.8 km at factor 0.7, as the model-anchored baseline is approximately $2.3\times$ the demand-clearing minimum.

4.2.2 Initial Backlog Sensitivity

Table D.1 varies the initial backlog fraction across four levels (5–35%) to test whether the budget insensitivity result depends on the 15% baseline starting condition.

Results are summarised in Table D.1 (Appendix D).

S0, S1, and S2 are identical at every initial backlog level from 5% to 35% (max S1–S0 divergence <0.001 km). The scenario ordering (S6 best, S4 worst) holds at all levels. The system converges to a stable equilibrium independent of starting state, confirming deterioration rate, not initial backlog, is the binding constraint.

S3 improves backlog to 14.45 km (plastic's longer TTL). S4 is the worst backlog scenario (19.24 km): laterite's short TTL drives deterioration faster than rehabilitation clears it. Despite lowest construction-phase CO₂ (4,440 t), S4's high use-phase CO₂ (12,067 t from laterite's rougher surface) gives a whole-life total of 16,507 t. Under priority-based allocation S4 worsens to 20.22 km, crossing the 20% crisis threshold.

S5 (ODM=0.7) reduces backlog to 12.60 km, cutting cost by 25.1% and construction CO₂ by 25.0%. It is the only single lever that improves all three metrics simultaneously.

S6 achieves the best outcome on all metrics: 10.68 km backlog, £5.26M cost, and 16,182 t whole-life CO₂ (9.7% below S0).

4.3 Priority-Based Allocation Sensitivity

To test whether the laterite inversion is an allocation artefact or a genuine material property, all seven scenarios were re-run under priority-based allocation, where rehabilitation effort is directed proportionally to each material's current deterioration rate (Eq. (14)) rather than its fixed network share. Table 4.7 reports year-20 backlog under both rules.

Table 4.7: Priority-based allocation sensitivity: year-20 backlog (km). Positive Δ = priority allocation is worse.

Scenario	Proportional (km)	Priority (km)	Δ (km)	Interpretation
S0 Baseline	17.054	17.625	+0.571	Marginal worsening
S1 Low-budget	17.054	17.625	+0.571	Identical to S0
S2 High-budget	17.054	17.625	+0.571	Identical to S0
S3 Plastic-push	14.449	15.301	+0.852	Priority worsens plastic TTL benefit
S4 Laterite-heavy	19.244	20.217	+0.973	Inversion strengthens
S5 Strong enf.	12.597	13.055	+0.458	Marginal worsening
S6 Combined	10.683	11.269	+0.586	S6 remains best scenario

Three findings emerge. First, S4 worsens under priority allocation (20.22 km vs 19.24 km): directing 76% of effort to laterite cannot overcome the TTL constraint, confirming the inversion is a material property not an allocation artefact. Second, priority allocation is marginally worse than proportional in all scenarios, as it directs effort to faster-deteriorating (cheaper) materials at the expense of longer-lived ones; condition-severity allocation is not automatically superior when material durability varies. Third, S6 remains best (11.27 km), confirming enforcement dominates allocation effects. Linard (2009) and Shiboub and Assaf (2022) discuss SD allocation rules but neither identifies material durability heterogeneity as a moderating variable.

A complementary TTL sensitivity analysis tests whether the laterite inversion is robust to uncertainty in the calibrated TTL value. The analysis re-runs S0 and S4 across laterite TTL values from 2.0 to 5.5 yr (the calibrated value is 3.33 yr; the upper plausible range in the tropical sealed-road literature is approximately 4.5–5.0 yr). Table 4.8 reports year-20 backlog for S0 and S4 at each level.

Table 4.8: TTL laterite sensitivity: inversion persists to TTL 4.5 yr, breakeven at 5.0 yr.

TTL laterite (yr)	S0 yr-20 (km)	S4 yr-20 (km)	S4 – S0 (km)	Inversion persists?
3.33	17.06	19.26	+2.20	Yes (calibrated)
4.0	16.68	17.87	+1.20	Yes
4.5	16.40	16.96	+0.56	Yes
5.0	16.14	16.14	0.00	Breakeven
5.5	15.88	15.39	–0.49	No

The inversion persists across the plausible TTL range (2.0–4.5 yr) and disappears only at TTL = 5.0 yr, the asphalt benchmark. Since the tropical sealed-road literature (Pinard et al., 2003; Kenya Ministry of Transport, Infrastructure, Housing and Urban Development, 2017) addresses low-volume roads with longer service lives than federal corridors, this threshold lies at or above the upper plausible bound. If local TTL meets the asphalt benchmark, the network risk disappears; the VoI analysis

below monetises what resolving this uncertainty is worth.

Value-of-information analysis for laterite TTL

The TTL sensitivity table identifies a threshold; a value-of-information (VoI) analysis converts it into a decision about whether to invest in corridor-specific TTL measurement. The decision is binary: adopt the laterite-heavy strategy (S4) or the baseline (S0). The outcome metric is year-20 backlog cost, valued at the weighted unit rehabilitation cost (£19,851/km) over the 100 km network. Two quantities are reported: (i) the *cost of the wrong decision* at a given TTL, and (ii) the Expected Value of Perfect Information (EVPI), the maximum a rational agency would pay for a measurement campaign that resolves TTL uncertainty entirely.

Table 4.9: Value-of-information analysis for laterite TTL. EVPI = maximum rational expenditure on corridor-specific TTL measurement. Cost of error at calibrated TTL: selecting S4 over S0 when TTL = 3.33 yr costs £43,672 in additional backlog rehabilitation (£437/km of network).

Prior TTL distribution	P(TTL $\geq$ 5 yr)	Optimal decision	EVPI (100 km corridor)	EVPI per corridor-km
Point estimate (3.33 yr)	0%	S0 (baseline)	£0	—
MC range U(2.78, 4.16 yr)	0%	S0 (baseline)	£0	£0
Moderate U(2.0, 7.0 yr)	40%	S0 (baseline)	£7,773	£78
Optimistic U(2.0, 9.0 yr)	57%	S4 (laterite)	£17,942	£179

Within the Monte Carlo calibration range (TTL 2.78–4.16 yr), S0 dominates across all samples and EVPI is zero: no measurement campaign can change the recommendation. EVPI becomes material only when an agency assigns substantial prior probability to TTL $\geq$ 5 yr. Under a moderate prior U(2.0, 7.0 yr) (P(TTL $\geq$ 5) $\approx$ 40%), EVPI reaches £78/km, the rational ceiling for a field TTL campaign. If field evidence instead confirms TTL $\approx$ 3.33 yr, the cost of mistakenly adopting S4 is £437/km.

4.4 Network-Level Monte Carlo

To test robustness across the plausible parameter space, a network-level Monte Carlo varies ODM (0.6–1.4) and initial backlog (5–35%) jointly across $N = 1,000$ LHS samples (seed 42), evaluating S0, S5, and S6 at each point. Table 4.10 reports distributions.

Table 4.10: Network-level Monte Carlo: year-20 backlog distribution (km) for S0, S5, S6 across $N = 1,000$ LHS samples. S6 < S5 < S0 in all 1,000 samples.

Percentile	S0 backlog (km)	S5 backlog (km)	S6 backlog (km)	S5 saving vs S0	S6 saving vs S5
p5	11.65	8.47	7.06	3.18	1.41
p25	14.14	10.35	8.65	3.79	1.70
p50	17.05	12.60	10.55	4.45	2.05
p75	19.79	14.74	12.39	5.05	2.35
p95	21.85	16.37	13.80	5.48	2.57
<i>S5 < S0 in 1,000/1,000 samples (100%); S6 < S5 in 1,000/1,000 samples (100%)</i>					

S6 < S5 < S0 in all 1,000 samples on both backlog and whole-life CO₂. Median S6 backlog (10.55 km) is 38% below S0 (17.05 km); S6's IQR is 34% narrower than S0's. Median S6 CO₂

(15,777 t) is 12.3% below S0 (17,998 t). The narrower IQR is analytically significant: it means the combined policy reduces not only expected backlog but also the system’s sensitivity to enforcement effectiveness and initial conditions. A policy that performs well on average but with wide variance is operationally riskier than one with a tighter outcome distribution.

4.5 MCDA Robustness Check

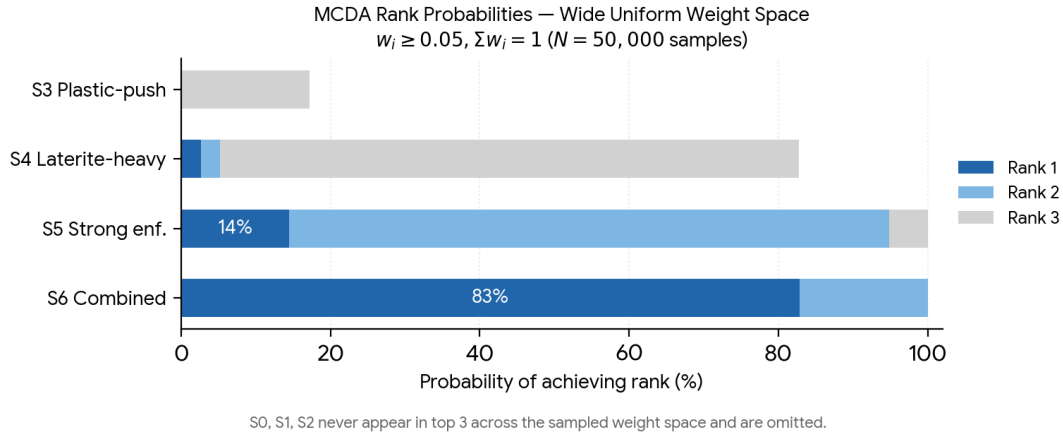


Figure 4.8: MCDA rank probabilities under a wide uniform weight space ($w_i \geq 0.05, \sum w_i = 1, N = 50,000$ Dirichlet samples). S6 ranks first in 82.9% of samples; S5 ranks first in 14.4%; S4 in 2.7% (when CO₂ or cost weight dominates). S0, S1, S2, and S3 never achieve rank 1 across the sampled space. As a boundary check, the analysis was also run with each weight constrained to a narrow band around its Dirichlet mean: S6 retained first rank in all cases, confirming the ordering is not sensitive to the shape of the weight distribution.

S6 ranks first in 100% of the 10,000 valid weight samples within the constrained weight space, confirming the scenario ordering is robust to priority reassignment within the tested range. Under the wider Dirichlet weight space (Figure 4.8), S4 achieves rank 1 in only 2.7% of samples, exclusively when CO₂ weight exceeds 0.6. This quantifies the trade-off: an agency would need to place negligible value on backlog reduction to prefer S4 over S6, a position difficult to defend operationally.

4.6 Budget-Equivalent Analysis of Enforcement Benefits

Table 4.11: Budget-equivalent interpretation of enforcement benefits.

Metric	S5 Strong enforcement	S6 Combined policy
Backlog reduction vs S0 (km)	4.46	6.37
Annual budget-equivalent (GBP/yr)	88,481	126,477
20-year budget-equivalent (GBP)	1,769,615	2,529,537
Construction-phase CO ₂ saving vs S0 (t)	1,683	1,800
Whole-life CO ₂ saving vs S0 (t)	1,074	1,747
Agency cost saving vs S0 (GBP)	1,695,759	1,525,281

Strong enforcement is equivalent to an additional £88,481/year in backlog benefit. A WIM enforcement station costs approximately £120,000–320,000 installed (World Road Association (PIARC), 2022; Torres Martínez et al., 2018). At the model’s annual saving of £84,788/yr (Table 4.11), the payback is 1.4–3.8 years from agency savings alone, before fines or user-cost benefits.

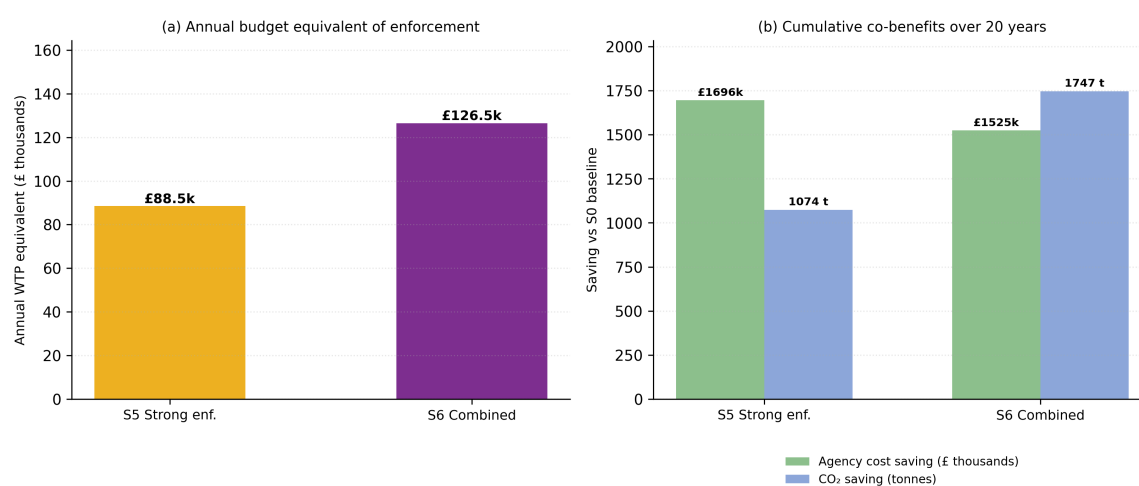


Figure 4.9: Annual budget-equivalent of enforcement (left) and cumulative co-benefits vs S0 baseline (right). CO₂ savings shown are construction-phase only; whole-life savings are approximately 6% (S5) and 9.7% (S6).

5 Discussion

5.1 Asset-Level Material Performance (RQ1)

The asset-level results directly challenge the use-phase dominance framing of Santero et al. (2011b). Asphalt's whole-life total (335.4 t) splits 58% construction-phase and 42% use-phase, a reversal of the temperate-climate pattern (Figure 4.2). Each avoided overlay removes a discrete construction event, so maintenance-reducing interventions deliver larger carbon co-benefits than Santero's framework predicts.

The CI-dependent RRC model (Eq. (7)) reveals that laterite's use-phase penalty is structural: shorter reseal cycles sustain lower in-service condition (mean CI 66.7 vs 68.6 for asphalt), generating rolling-resistance CO₂ that fixed-RRC assumptions would miss. Plastic's higher CI (71.8) offsets its construction emissions, retaining an 81.5 t advantage over asphalt. At a 25-year horizon, NPV ranking is unchanged at 12% discount rate, but plastic's undiscounted advantage grows from £3,400 to £37,000 and its CO₂ advantage doubles to 129 t. Agencies applying high discount rates will systematically undervalue plastic relative to its whole-life performance.

Laterite's CO₂ advantage is robust, but its cost advantage is strong rather than absolute: its 95th-percentile NPV (£135,945) overlaps with both other materials. The Sobol analysis shows CO₂ uncertainty is driven by the $k_{\text{clim}} \times \alpha$ interaction rather than either parameter alone, meaning agencies facing both high tropical exposure and severe overloading face compound uncertainty that OAT would understate. This interaction has a policy implication beyond sensitivity analysis: overload enforcement attenuates the α -driven damage amplification component, so by reducing axle loads, enforcement does not merely slow deterioration but compresses the variance in CO₂ trajectories under uncertain climate conditions. An agency that reduces overloading therefore manages both expected CO₂ and CO₂ uncertainty simultaneously. This is part of why S5 and S6 produce narrower outcome distributions in the network Monte Carlo: enforcement acts as a variance-reduction mechanism, a result OAT cannot reveal.

Santero et al. (2011a) found that maintenance schedule assumptions dominate pavement LCA sensitivity more than material emission factors. This study extends that finding: under a fixed 10-year overlay schedule, plastic appears more expensive throughout, but under the endogenous condition-driven schedule, plastic reaches cost parity at year 22 because its longer TTL delays the second overlay. The schedule assumption determines the qualitative conclusion, not merely its magnitude. This is precisely the kind of result that justifies condition-driven simulation over static LCA: the decision between materials depends on how maintenance is triggered, and only a model that generates maintenance endogenously can detect this dependence.

5.2 Network-Level Findings and Their Interpretation

The complete identity of S0, S1, and S2 is the most structurally significant result. As established in Section 4.2, the budget is non-binding because rehabilitation capacity structurally exceeds deterioration inflow under the model-anchored budget; the binding threshold (factor 0.41) lies well below

all tested scenarios.

This directly challenges Foster and Pushak (2011)’s framing of Nigeria’s road problem as a structural funding deficit. The model agrees a threshold exists (factor 0.41) below which backlog diverges, but above it, budget variation is irrelevant. Linard (2009) and Ruiz and Guevara (2020a) identify similar deterioration-dominated dynamics but do not isolate a mechanism that shifts the constraint. This study does: enforcement reduces ODM, lowers deterioration inflow, and expands the range within which capacity exceeds demand. The policy problem is reframed from funding rehabilitation to reducing demand for it. This distinction matters practically because it redirects attention from treasury negotiations to operational enforcement infrastructure, a fundamentally different institutional pathway that does not require increased appropriations.

There is also a methodological implication for LCA. The Santero et al. (2011a) critique of fixed maintenance schedules in LCA implicitly assumes that a budget-constrained agency would deviate from those schedules when funds are insufficient. The budget non-binding result shows that, for the parameterisation used here, the modelled agency always has sufficient capacity to execute its condition-triggered maintenance schedule. The fixed-schedule assumption in static LCA is therefore not unreasonable for corridors above the binding threshold — the SD model provides an unexpected structural justification for the LCA convention it was designed to supersede. This does not rehabilitate static LCA, because it still cannot detect the laterite inversion, but it clarifies the specific domain in which the two approaches agree.

The one-way coupling addresses RQ2 by propagating material-specific TTL values, unit costs, and CO₂ burdens into the SD model, making engineering differences traceable to network-level backlog outcomes and addressing the gap noted by Talaghat et al. (2024). The translation is necessarily lossy: condition-band logic is compressed into a single TTL scalar per material. The compression is quantifiable: inter-overlay interval CVs are 37.7% (asphalt), 45.1% (laterite), and 36.2% (plastic), so the mean TTL captures central tendency but suppresses within-material variability; the TTL sensitivity analysis (Table 4.8) partially compensates. Under Kritzinger et al. (2018)’s taxonomy the framework sits closer to a digital model than a full digital twin; ‘digital-twin-enabled’ denotes architectural intent for future data integration.

5.3 Policy Implications

Figure 5.1 summarises all seven scenarios on four metrics simultaneously, making trade-offs visible that Table 4.5 conveys only numerically.

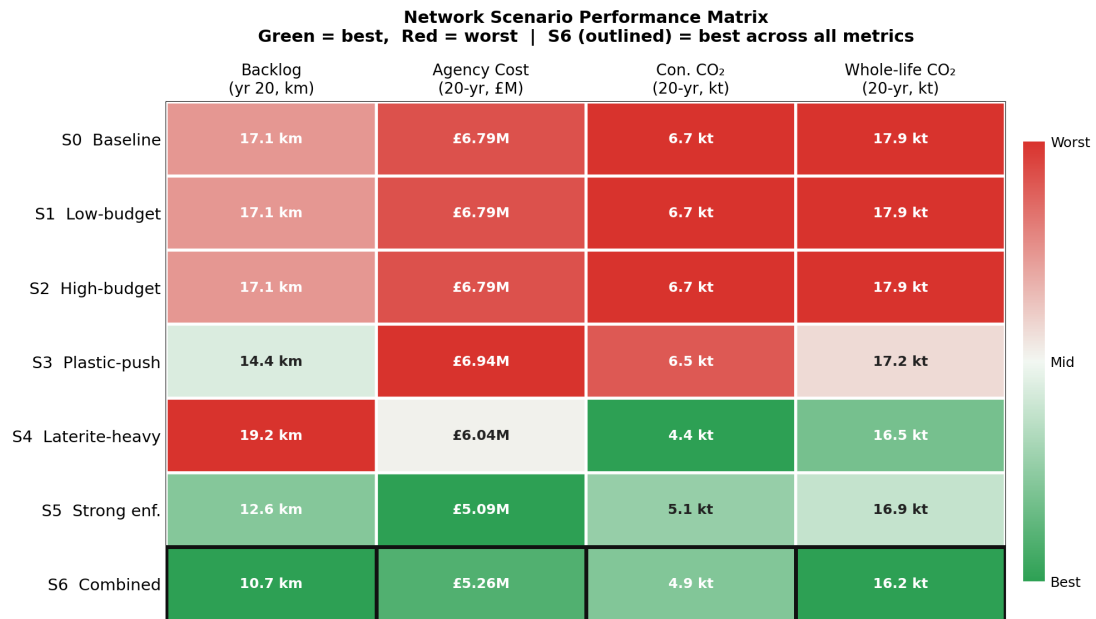


Figure 5.1: Network scenario performance matrix. Each cell shows the raw metric value; colour indicates relative performance across all seven scenarios (green = best, red = worst, normalised within each metric). S6 (outlined) achieves the best or near-best outcome on all four metrics. S0, S1 and S2 are identical, confirming the budget is non-binding. S4’s green CO₂ cells alongside its red backlog cell show the trade-off that makes asset-level LCA an insufficient basis for network strategy.

Three policy implications follow from the results.

First, overload enforcement (S5) is the only lever simultaneously reducing backlog, cost, and CO₂. Ruiz and Guevara (2020b) show that demand-side strategies can outperform corrective rehabilitation at network scale; the present study mechanistically explains why: enforcement reduces ODM, lowers effective deterioration rate, and reduces rehabilitation demand, expanding the range within which capacity exceeds demand. The budget-equivalent framing makes this concrete: enforcement delivers what £88,481/yr of additional spending cannot, because under a deterioration-dominated regime additional budget has no productive outlet.

Second, S4 reveals that asset-level LCA is insufficient for network strategy. Laterite’s short TTL (3.33 yr) generates deterioration faster than the rehabilitation budget can clear at scale; the inversion persists under priority-based allocation and across the tested TTL range (Tables 4.7 and 4.8), disappearing only at the asphalt benchmark (5.0 yr). The VoI analysis (Table 4.9) converts this threshold into a measurement budget: a corridor survey is worth up to £78/km at 40% prior probability of $TTL \geq 5$ yr; the cost of mistakenly adopting S4 is £437/km. This finding has implications beyond Nigeria: any context where a low-cost material has a substantially shorter service life than the network’s dominant surfacing is susceptible to the same inversion. The mechanism is generic to the coupling architecture, not specific to laterite.

Third, enforcement and material substitution are complementary: enforcement reduces deterioration inflow while plastic’s longer TTL slows its accumulation. The network Monte Carlo (Table 4.10) confirms $S6 < S5 < S0$ in all 1,000 samples (ODM 0.6–1.4, initial backlog 5–35%). Median S6 backlog is 38% below S0 with a 34% narrower IQR. Adopting plastic without enforcement leaves half the improvement on the table.

These are engineering findings within the model space. Practical implementation depends on weigh-

in-motion infrastructure, enforcement capacity, credible sanctions, and political commitment.

5.4 Limitations and Contextualisation

Results fall into three tiers: *robust* (CO₂ ranking, $S6 > S5 > S0$ in all MC draws); *strong but model-dependent* (budget non-binding, laterite inversion); and *indicative* (whole-life network CO₂). Three limitations have identifiable bias directions, all of which work against the enforcement and plastic findings. First, the modelled HGV share (15.9%) is at the lower end of the 13.7–30.5% inter-corridor range; a higher share would amplify deterioration and strengthen the enforcement benefit, making the reported S5 and S6 advantages conservative. Second, the linear CI-to-IRI approximation understates the roughness penalty at high IRI relative to the log-linear AASHTO relationship, which underestimates laterite’s use-phase CO₂ penalty at low condition and makes the laterite network inversion conservative in magnitude. Third, the fixed-RRC assumption in the SD model understates whole-life CO₂ for high-backlog scenarios, making the enforcement CO₂ advantage appear smaller than it would under a full-roughness model. All three biases mean the reported advantages of S5 and S6 are likely underestimates under more realistic parameterisation. Further limitations include CO₂-only environmental scope and stylised cross-corridor parameterisation.

6 Conclusions and Recommendations

6.1 Summary of Findings

This study developed a two-layer digital-twin-enabled LCA framework to evaluate three pavement materials for Nigerian road conditions. The key findings against each research question are summarised below.

RQ1. Laterite has the lowest NPV (£79,688) and construction CO₂ (44.7 t). Whole-life hierarchy is maintained (laterite 236.7 t, plastic 253.9 t, asphalt 335.4 t) after CI-dependent RRC; asphalt's 58% construction-phase dominance contradicts Santero's use-phase-dominant prediction. Laterite's CO₂ advantage is robust; its cost advantage is strong but not absolute.

RQ2. One-way coupling enables the laterite inversion to be identified: laterite, the best asset-level performer, becomes the highest-risk network scenario due to its short TTL. This finding is inaccessible from asset-level LCA alone and constitutes the clearest contribution of the coupling architecture.

RQ3. The modelled budget constraint is non-binding within the tested range (factors 0.41–1.30, initial backlogs 5–35%), a structural consequence of the endogenous budget construction rather than an empirical claim about Nigerian agencies. Enforcement (S5) is the only lever simultaneously reducing backlog, cost, and CO₂. S6 (combined policy) is the preferred scenario under MCDA and across all 1,000 network-level Monte Carlo samples.

6.2 Policy Recommendations

Immediate action: strengthen axle-load enforcement. Within the model space this is the only lever simultaneously reducing backlog, cost, and CO₂, delivering what £88,481/yr of additional spending cannot. A fixed WIM enforcement station costs approximately £120,000–320,000 installed (World Road Association (PIARC), 2022), implying a simple payback of 1.4–3.8 years from agency cost savings alone (Table 4.11) before enforcement revenue is included. Implementation further requires credible sanctions and political commitment to regulate freight operators.

Conditional action: expand laterite only where local TTL evidence supports it. The network risk disappears when laterite TTL meets the asphalt benchmark (≈ 5 yr; Table 4.8). A VoI analysis (Table 4.9) shows that a corridor TTL measurement campaign is economically justified (EVPI £78/km) only when the agency assigns at least 40% prior probability to $TTL \geq 5$ yr; otherwise the calibrated TTL of 3.33 yr should drive the decision. Agencies should pair any adoption with drainage investment and enforcement.

Strategic action: adopt plastic-modified asphalt where CO₂ reduction has policy value. The NPV premium persists at 12% discount rate, but plastic's CO₂ advantage grows with horizon (63.8 t at year 20; 129 t at year 25) and its undiscounted cost advantage emerges by year 22.

6.3 Decision Framework for Material Selection

Table 6.1 synthesises the dissertation's findings into an operational decision guide for road agencies. Three binary conditions govern the recommendation: whether corridor-specific TTL evidence is available, whether CO₂ reduction is a procurement or policy objective, and whether planning operates at a low discount rate (below approximately 8%, where future cost avoidance carries material weight) or the standard Nigerian rate (12%).

Table 6.1: Material selection decision framework. TTL evidence = corridor-specific laterite service-life data meeting or exceeding the asphalt benchmark (≥ 5 yr); CO₂ priority = CO₂ reduction carries direct procurement or policy weight; Low discount rate = below $\approx 8\%$. Recommendations are conditional on the modelled parameterisation; agencies should adjust thresholds using local data.

TTL evidence (≥ 5 yr)?	CO ₂ priority?	Low discount rate?	Recommended material	Primary reason
No	No	No	Laterite (with enforcement)	Lowest NPV; short TTL risk mitigated by enforcement
No	No	Yes	Plastic-modified asphalt	Undiscounted advantage grows with horizon
No	Yes	No	Plastic-modified asphalt	CO ₂ advantage robust; grows to 129 t at yr 25
No	Yes	Yes	Plastic-modified asphalt	Strongest combined CO ₂ and cost case
Yes	No	No	Laterite	TTL evidence removes network inversion risk; lowest cost
Yes	No	Yes	Laterite	Strongest cost case; network risk resolved
Yes	Yes	No	Laterite or plastic (tied)	CO ₂ advantages differ in form (construction vs use-phase)
Yes	Yes	Yes	Plastic-modified asphalt	Longest service cycles sustain best condition; growing CO ₂ advantage

All eight scenarios assume overload enforcement is in place. Without enforcement, laterite's short TTL drives network deterioration regardless of TTL evidence, and the S4 inversion applies in all rows where TTL evidence is absent.

6.4 Research Contributions Summary

Table 6.2 maps each research question to its principal finding and the resulting policy implication.

Table 6.2: Research questions, key findings, and policy implications.

RQ	Key Finding	Policy Implication
RQ1 <i>Asset-level</i>	Sealed laterite has the lowest NPV (£79,688) and construction-phase CO ₂ (44.7 t). Whole-life hierarchy is maintained (laterite 236.7 t < plastic 253.9 t < asphalt 335.4 t). Laterite CO ₂ advantage is robust across the full Monte Carlo parameter space.	Adopt laterite where drainage and borrow-pit quality can be controlled. Commission a corridor TTL measurement campaign only if $P(TTL \geq 5 \text{ yr}) \geq 40\%$ (EVPI = £78/km). Adopt plastic-modified asphalt where CO ₂ reduction carries direct policy or procurement weight; its advantage grows to 129 t at year 25.
RQ2 <i>Network-level</i>	Laterite's asset-level advantage inverts at network scale: the laterite-heavy scenario (S4) produces the worst backlog (19.2 km vs S0 baseline 17.1 km) because its short TTL generates a deterioration inflow the rehabilitation budget cannot clear. This result is inaccessible from asset-level LCA alone.	Require network-level analysis before scaling any material to a dominant network share. The one-way coupled framework provides a low-data template replicable on other Nigerian or Sub-Saharan African corridors without requiring sensor infrastructure.
RQ3 <i>Scenario</i>	Budget variation of $\pm 30\%$ produces no backlog change (non-binding above factor 0.41). Enforcement (S5) is the only lever simultaneously reducing backlog, cost, and CO ₂ . S6 (combined policy) ranks first in 82.9% of wide weight-space samples and in all 1,000 network Monte Carlo draws.	Prioritise axle-load enforcement over budget increases (WIM payback 1.4–3.8 yr from agency savings alone). Plan enforcement and plastic-modified asphalt adoption jointly; S6 outperforms either lever alone. The binding constraint is deterioration rate, not expenditure level.

6.5 Limitations and Future Work

Principal limitations are calibration to literature targets, CO₂-only environmental scope, endogenous budget anchoring, and a stylised 100 km aggregate. Future work should prioritise: (1) field calibration of deterioration rates and TTL values against Nigerian condition surveys, which would convert the model-space findings into corridor-specific predictions; (2) extension of the CI-dependent RRC model to the network layer by coupling CI distributions to the Vensim km-stocks, eliminating the fixed-RRC limitation that currently makes network-level CO₂ rankings indicative rather than robust; (3) spatial network representation to capture corridor heterogeneity in subgrade quality and traffic; and (4) weigh-in-motion data integration to close the loop between enforcement monitoring and deterioration modelling (Kritzinger et al., 2018).

6.6 Concluding Statement

Under the adopted parameterisation, maintenance performance is more sensitive to deterioration-rate management than to budget variation. Above the identified threshold, additional funding produces no improvement; enforcement and material strategy do. This conclusion requires the full two-layer architecture: neither asset-level nor network-level analysis alone can diagnose it.

The dissertation changes the state of knowledge in three respects. First, it demonstrates that the best asset-level material can produce the worst network outcome, a result absent from the existing pavement LCA literature. Second, it provides the first quantitative evidence that budget is non-binding for a Nigerian federal corridor parameterisation, reframing the policy problem from expenditure to deterioration management. Third, it shows that climate and loading interact multiplicatively on CO₂ outcomes, meaning that one-at-a-time sensitivity methods systematically understate carbon uncertainty in tropical overloading contexts.

Bibliography

- O. Ademila. Engineering evaluation of lateritic soils of failed highway sections in southwestern Nigeria. *Geosciences Research*, 2(3), 2017.
- C.C. Akudo and G.O. Aigbadon. Geotechnical appraisal of failed sections of some highway pavements in parts of northcentral Nigeria. *World Journal of Engineering*, 2024.
- B. Alade and O. Amu. Comparison of the geotechnical properties of lateritic soils of borrow pits for road construction in Ado Ekiti. *ABUAD Journal of Engineering Research and Development*, 8(2):108–120, 2025.
- H. AzariJafari, A. Yahia, and M.B. Amor. Life cycle assessment of pavements: reviewing research challenges and opportunities. *Journal of Cleaner Production*, 112:2187–2197, 2016.
- Y. Barlas. Formal aspects of model validity and validation in system dynamics. *System Dynamics Review*, 12(3):183–210, 1996.
- V.S. de F. Bello et al. Framework for bridge management systems using digital twins. In *EU-ROSTRUCT 2021*, Cham, 2022. Springer.
- Karim Chatti and Imen Zaabar. Estimating the effects of pavement condition on vehicle operating costs. National Cooperative Highway Research Program Report NCHRP Report 720, Transportation Research Board, Washington, D.C., 2012.
- S.O. Daramola, S. Malomo, and Y.A. Asiwaju-Bello. Premature failure of a major highway in southwestern Nigeria: the case of Ipele–Isua highway. *International Journal of Geo-Engineering*, 9:28, 2018.
- Ekpenyong Ekpo Efiom, A. S. P. Abu, and K. Z. Cinfwat. Comparative study of the road roughness measurement of RoadLab Pro and Roadroid applications for IRI data collection in Nigeria. *The International Journal of Engineering and Science*, 10(5):14–19, 2021.
- V. Foster and N. Pushak. *Nigeria's Infrastructure: A Continental Perspective*. World Bank, Washington DC, 2011.
- Jeremy Gregory, Arash Noshadravan, Elsa A. Olivetti, and Randolph Kirchain. A methodology for robust comparative life cycle assessments incorporating uncertainty. *Environmental Science & Technology*, 50(12):6397–6405, 2016. doi: 10.1021/acs.est.5b04969.
- J.T. Harvey, J. Meijer, and A. Kendall. Pavement life cycle assessment framework. Technical Report FHWA-HIF-15-001, Federal Highway Administration, Washington DC, 2016.
- IPCC. 2006 IPCC guidelines for national greenhouse gas inventories, volume 2: Energy. Technical report, Institute for Global Environmental Strategies, Hayama, Japan, 2006.
- ISO. ISO 14040: Environmental management — life cycle assessment — principles and framework, 2006a.

- ISO. ISO 14044: Environmental management — life cycle assessment — requirements and guidelines, 2006b.
- Z. Jwaida, A. Dulaimi, M.A. Othuman Mydin, Y.O. Özkılıç, R. Putra Jaya, and A. Ameen. The use of waste polymers in asphalt mixtures: bibliometric analysis and systematic review. *Journal of Composites Science*, 7(10):415, 2023.
- Kenya Ministry of Transport, Infrastructure, Housing and Urban Development. *Pavement Design Guidelines, PDG1: Pavement Design Guideline for Low Volume Sealed Roads*. MOTI, Nairobi, 2017.
- W. Kritzing, M. Karner, G. Traar, J. Henjes, and W. Sihn. Digital twin in manufacturing: a categorical literature review and classification. *IFAC-PapersOnLine*, 51(11):1016–1022, 2018.
- K.T. Linard. Application of system dynamics to unsealed road maintenance management. In *Proceedings of the 27th International Conference of the System Dynamics Society*, 2009.
- K.T. Linard. A system dynamics modelling approach to gravel road maintenance management. In *Proceedings of the 24th ARRB Conference*, Melbourne, Australia, 2010.
- C. Liu, P. Zhang, and X. Xu. Literature review of digital twin technologies for civil infrastructure. *Journal of Infrastructure Intelligence and Resilience*, 2(3):100050, 2023.
- Arash Noshadravan, Margaret Wildnauer, Jeremy Gregory, and Randolph Kirchain. Comparative pavement life cycle assessment with parameter uncertainty. *Transportation Research Part D: Transport and Environment*, 25:131–138, 2013. doi: 10.1016/j.trd.2013.10.002.
- O.J. Oyekanmi, C.C. Ibe, O.T. Ebiringa, and E.A. Ejem. Analysis of the extent of overloading on the Nigerian highways. *International Journal of Transportation Engineering and Technology*, 6(1):22–29, 2020. doi: 10.11648/j.ijtet.20200601.14.
- J. Pan et al. A comprehensive review of applications and environmental risks of waste plastics in asphalt pavements. *Materials*, 18(15):3441, 2025.
- M.I. Pinard, S.D. Ellis, C-H. Eriksson, R. Johansen, T. Toole, R. Beger, M.E. Gumbie, H.J.S. Lotter, and A.R. Quimby. *SADC Guideline on Low-Volume Sealed Roads*. Southern African Development Community, Gaborone, 2003.
- Saheed Olanrewaju Popoola et al. Impact of pavement roughness on traffic safety under heterogeneous traffic conditions. *Nigerian Journal of Technological Development*, 17(1):13–21, 2020. doi: 10.4314/njtd.v17i1.1.
- Road Sector Development Team. Configuration and calibration of HDM-4 to Nigerian conditions. Draft final report, Federal Government of Nigeria, Federal Ministry of Works, 2014.
- A. Ruiz and J. Guevara. Environmental and economic impacts of road infrastructure development: dynamic considerations and policies. *Journal of Management in Engineering*, 36(3):04020006, 2020a.
- A. Ruiz and J. Guevara. Sustainable decision-making in road development: analysis of road preservation policies. *Sustainability*, 12(3):872, 2020b.

- Andrea Saltelli, Paola Annoni, Ivano Azzini, Francesca Campolongo, Marco Ratto, and Stefano Tarantola. Variance based sensitivity analysis of model output: Design and estimator for the total sensitivity index. *Computer Physics Communications*, 181(2):259–270, 2010. doi: 10.1016/j.cpc.2009.09.018.
- N.J. Santero, E. Masanet, and A. Horvath. Life-cycle assessment of pavements. Part I: critical review. *Resources, Conservation and Recycling*, 55(9–10):801–809, 2011a.
- N.J. Santero, E. Masanet, and A. Horvath. Life-cycle assessment of pavements. Part II: filling the research gaps. *Resources, Conservation and Recycling*, 55(9–10):810–818, 2011b.
- I. Shiboub and G.J. Assaf. System dynamic model for sustainable road rehabilitation integrating technical, economic, and environmental considerations. *Journal of Management in Engineering*, 38(5):04022041, 2022.
- J.D. Sterman. *Business Dynamics: Systems Thinking and Modeling for a Complex World*. McGraw-Hill, Boston, 2000.
- M.A. Talaghat, A. Golroo, A. Kharbouch, M. Rasti, R. Heikkilä, and R. Jurva. Digital twin technology for road pavement. *Automation in Construction*, 168:105826, 2024.
- A.J. Torres Martínez, S.-O. Josa, F. Magrinya, and J.-M. Gauthier. Cost-effectiveness of enforcing axle-load regulations: The Douala–N’Djamena corridor in Sub-Saharan Africa. *Transportation Research Part A: Policy and Practice*, 107:216–228, 2018. doi: 10.1016/j.tra.2016.11.005.
- Transport Research Laboratory. Overseas road note 31: A guide to the structural design of bitumen-surfaced roads in tropical and sub-tropical countries. Technical report, Transport Research Laboratory, Crowthorne, UK, 1993.
- L. Trupia, T. Parry, L.C. Neves, and D. Lo Presti. Rolling resistance contribution to a road pavement life cycle carbon footprint analysis. *The International Journal of Life Cycle Assessment*, 22(6): 972–985, 2017.
- H.K. Ugwuanyi and K.C. Onyelowe. Suitability of Olokoro and Amaoba lateritic soil as pavement construction materials. *Nigerian Journal of Technology*, 38(3):549–556, 2019.
- World Bank. Project appraisal document on a proposed credit to the federal republic of nigeria for the federal roads development project, 2008. Reports Lagos–Ibadan Expressway traffic conditions including about 16,964 vehicles/day in each direction, 13.7% heavy vehicles, and an assumed 4% annual growth rate.
- World Road Association (PIARC). Overweight vehicles: Impact on road infrastructure and safety. Technical Report 2022SP01EN, PIARC Special Project, 2022.
- F. Xu, Y. Zhao, and K. Li. Using waste plastics as asphalt modifier: a review. *Materials*, 15(1):110, 2022.
- J.A. Yakubu, S.S. Daku, R.B. Gusikit, E.D. Emmanuel, and M.C. Mangai. Causes of failure of Nigerian roads: a review. *World Journal of Advanced Engineering Technology and Sciences*, 8(2):217–223, 2023.

A Parameter Register

This appendix records the final model parameters used in the dissertation, based directly on the exported files `config_used.json` and `vensim_params.csv`. Parameters are grouped into: (i) general model settings, (ii) traffic, climate, and condition inputs, (iii) deterioration and maintenance logic, (iv) cost and carbon parameters, and (v) MATLAB-to-Vensim exported material parameters.

A.1 General Model Settings

Table A.1: General model settings from `config_used.json`.

Parameter	Symbol	Value	Notes
Analysis horizon	T	20 years	Final model horizon
Time step	Δt	1 year	Annual timestep
Road length	—	1000 m	Functional unit length
Number of lanes	—	1	One design lane of a two-way traffic stream
Lane width	—	3.65 m	Standard lane width assumption
Area	—	3650 m ²	Derived from 1 km $\times$ 3.65 m
Base year	—	2023	Reference year in configuration
Exchange rate	—	1024.4 NGN/£	Stored in config file
Inverse exchange rate	—	0.00097618 £/NGN	Stored in config file
Discount rate	d	0.12	Used for NPV calculation

A.2 Traffic, Climate, and Condition Inputs

Table A.2: Traffic, climate, and condition inputs from `config_used.json`.

Parameter	Symbol	Value	Notes
AADT (two-way)	$AADT_0$	13,667 veh/day	Representative federal-road scenario baseline
Heavy vehicle share	p_{HGV}	0.159	15.9%
Traffic growth rate	g	0.04	4%/year
Lane split factor	—	0.5	Two-way traffic split
ESAs per HGV, no overload	—	1.6	Non-overloaded reference case
ESAs per HGV, overload	ESA_{HGV}	10.0	Overloaded baseline case
Loading mode	—	overload	Active baseline mode
Loading exponent	α	0.3	Used in traffic damage scaling
Reference annual ESAs	ESA_{ref}	634,531.476	Noload-basis reference (1.6 ESA/HGV); overload amplification expressed via $f_{traffic}$
Climate multiplier	k_{clim}	1.15	Humid tropical assumption
Initial condition index	CI_0	100	New-pavement starting value
Failure threshold	—	40	Reconstruction threshold
Condition cap	—	100	Upper CI bound

A.3 Deterioration Structure

Table A.3: Deterioration parameters from `config_used.json`.

Parameter	Symbol	Value	Notes
Good condition band	—	70–100	CI band
Mid condition band	—	55–70	CI band
Poor condition band	—	40–55	CI band
Very poor condition band	—	0–40	CI band
Good-band deterioration rate	r_{good}	2.2 CI pts/yr	Final calibrated value
Mid-band deterioration rate	r_{mid}	3.0 CI pts/yr	Final calibrated value
Poor-band deterioration rate	r_{poor}	4.2 CI pts/yr	Final calibrated value
Very poor-band deterioration rate	r_{vpoor}	6.0 CI pts/yr	Final calibrated value

A.4 Maintenance and Intervention Logic

Table A.4: Maintenance logic from `config_used.json`.

Parameter	Symbol	Value	Notes
Minor repair threshold	CI_{minor}	65	Trigger for minor repair
Overlay threshold	CI_{overlay}	55	Trigger for overlay / reseal
Reconstruction threshold	CI_{recon}	40	Trigger for reconstruction
Minor repair recovery	ΔCI_{minor}	+4	CI boost
Overlay recovery	$\Delta CI_{\text{overlay}}$	+28	CI boost
Reconstruction reset	—	100	Full condition reset
Minimum interval between minor repairs	—	3 years	Prevents unrealistic clustering
Minor repair area fraction	—	0.08	8% patched area assumption
Maintenance delay	—	0 years	No intervention delay applied

A.5 Cost Parameters

Table A.5: Cost parameters from `config_used.json`.

Parameter	Symbol	Value	Notes
Asphalt overlay base cost	—	9000 NGN/m ²	Baseline overlay cost
Asphalt overlay range	—	6000–12000 NGN/m ²	Sensitivity range
Reconstruction / overlay ratio (base)	—	2.5	Cost multiplier
Reconstruction / overlay ratio range	—	1.8–3.0	Sensitivity range
Laterite reseal ratio (base)	—	0.6	Relative to asphalt overlay
Laterite reseal ratio range	—	0.5–0.7	Sensitivity range
Laterite construction ratio	—	0.75	Relative to asphalt construction
Minor fraction of overlay (base)	—	0.10	Minor repair cost fraction
Minor fraction of overlay range	—	0.05–0.15	Sensitivity range
Plastic premium (base)	—	0.20	Relative to conventional asphalt
Plastic premium range	—	0.10–0.30	Sensitivity range

Table A.6: Material-specific intervention cost values from `config_used.json`.

Material	Construction	Minor repair	Overlay / reseal	Reconstruction
Asphalt	22,500 NGN/m ²	900 NGN/m ²	9000 NGN/m ²	22,500 NGN/m ²
Laterite	16,875 NGN/m ²	720 NGN/m ²	5400 NGN/m ²	16,875 NGN/m ²
Plastic-modified asphalt	27,000 NGN/m ²	1080 NGN/m ²	10,800 NGN/m ²	27,000 NGN/m ²

A.6 Carbon Parameters

Table A.7: Carbon and material-emissions parameters from `config_used.json`.

Parameter	Symbol	Value	Notes
Asphalt mix emission factor	—	0.055	Stored carbon parameter
Aggregate emission factor	—	0.01	Stored carbon parameter
Bitumen emission factor	—	0.4	Stored carbon parameter
Asphalt density	—	2400 kg/m ³	Material property
Aggregate density	—	2000 kg/m ³	Material property
Asphalt overlay thickness	—	0.05 m	Overlay depth assumption
Asphalt patch thickness	—	0.04 m	Patch depth assumption
Asphalt reconstruction thickness	—	0.1262 m	Reconstruction thickness assumption
Unbound reconstruction thickness	—	0.35 m	Reconstruction support layer assumption
Process uplift factor	—	1.2	Process-related carbon uplift
Patch area fraction	—	0.08	8% patching assumption
Seal binder application	—	1.2 kg/m ²	Laterite reseal parameter
Seal chips application	—	14 kg/m ²	Laterite reseal parameter

A.7 Calibration Targets

Table A.8: Calibration target windows from `config_used.json`.

Calibration item	Target window	Notes
Asphalt first overlay	8–12 years	Targeted by calibration
Asphalt first reconstruction	15–25 years	Targeted by calibration
Laterite first overlay / reseal	5–9 years	Targeted by calibration
Plastic first overlay	10–15 years	Targeted by calibration

A.8 Exported MATLAB-to-Vensim Parameters

Table A.9: Material-specific parameters exported to `vensim_params.csv`.

Material	ESA/HV	Growth	Mean maint. int.	Mean minor int.	Mean overlay int.	Service life	NPV (£)
Asphalt	10.0	0.04	2.4 yr	5.0 yr	5.0 yr	20 yr	99,872.50
Laterite	10.0	0.04	1.7143 yr	3.6667 yr	3.3333 yr	20 yr	79,687.93
Plastic-modified asphalt	10.0	0.04	3.0 yr	7.0 yr	7.0 yr	20 yr	110,691.65

Table A.10: Additional exported material results from `vensim_params.csv`.

Material	Cost per km-year (£)	Construction- phase CO ₂ per km-year (t)	20-yr construction- phase CO ₂ (t)	n_{minor}	n_{overlay}	n_{recon}
Asphalt	8857.06	9.7949	195.8981	3	3	0
Laterite	6895.49	2.2336	44.6725	4	4	0
Plastic-modified asphalt	8689.02	6.6056	132.1120	2	2	0

A.9 Use-Phase CO₂ Parameters

The following parameters govern the use-phase rolling-resistance CO₂ model (Eqs. (15) and 6).

Table A.11: Use-phase rolling-resistance CO₂ parameters. Sources: Trupia et al. (2017); Chatti & Zaabar (2012) NCHRP 720; IPCC (2006) Tier 1.

Parameter	Symbol / Variable	Value	Notes
RRC, conventional asphalt	RRC_{asp}	0.0120	Good condition, Chatti & Zaabar (2012)
RRC, plastic-modified asphalt	RRC_{pla}	0.0105	Stiffer mix, lower resistance
RRC, sealed laterite	RRC_{lat}	0.0165	Rougher sealed surface
Fuel factor, light vehicles	FF_{LV}	0.040 L/veh- km/unit-RRC	NCHRP 720
Fuel factor, HGV	FF_{HGV}	0.150 L/veh- km/unit-RRC	NCHRP 720
Fleet fuel factor	F_{mix}	0.0437 L/veh- km/unit-RRC	$(1 - 0.159) \times 0.040 + 0.159 \times 0.150$
Diesel emission factor	EF	2.64 kg CO ₂ /L	IPCC (2006) Tier 1; diesel-equivalent
Traffic growth rate	g	4%/yr	Same as main model
Lane split	λ	0.50	One design lane
Backlog exclusion	—	Applied	Backlog km excluded; see Section 3.3.4

Table A.12: Asset-level whole-life CO₂ summary. Construction-phase values from `vensim_params.csv`; use-phase values computed from Eq. (6) with condition-index-dependent RRC scaling (Eq. (7)), 4%/yr traffic growth.

Material	Construction CO ₂ (t)	Use-phase CO ₂ (t)	Whole-life CO ₂ (t)
Conventional asphalt	195.9	139.5	335.4
Sealed laterite	44.7	192.0	236.7
Plastic-modified asphalt	132.1	121.8	253.9

A.10 Notes

The values reported above are taken directly from the uploaded exports `config_used.json` and `vensim_params.csv`. Where the dissertation text reports rounded values in the main chapters, Appendix A should be treated as the exact machine-readable parameter record used for model execution and coupling.

B Key MATLAB Code Listings

This appendix provides the key MATLAB function files used in the asset-level digital-twin-enabled simulation. Four core functions are listed: `config_base.m` defines the baseline parameter structure; `simulate_section.m` orchestrates the annual time-step loop; `deterioration_step.m` computes condition loss per year; and `event_cost_emissions.m` assigns cost and construction-phase CO₂ to each maintenance event. Together these four functions constitute the central computational engine for the MATLAB deterioration, cost, and embodied-carbon results. Use-phase rolling-resistance CO₂ is computed separately via `rrc_annual_co2.m` (Eq. (6)); whole-life asset totals are the sum of the construction-phase output of these four functions and the use-phase output of that standalone function.

B.1 `rrc_condition_factor.m`: Condition-Dependent RRC Scaling

`rrc_condition_factor.m` returns the multiplicative factor $\phi(\text{CI})$ defined in Eq. (7). It is called once per annual time step in `simulate_section.m`, receiving the current-year CI before any maintenance intervention.

```
function rrc_factor = rrc_condition_factor(CI)
% RRC_CONDITION_FACTOR Condition-dependent RRC multiplier (phi(CI) scaling).
%
% Returns phi(CI) >= 1.0 scaling baseline RRC for in-service roughness.
%
% Method:
%   IRI(CI) = max(0, (100 - CI) / 15) [m/km]
%   rrc_factor = 1 + beta * IRI(CI)
%   beta = 0.014 (mixed-fleet: 84.1% LV @ 0.015 + 15.9% HGV @ 0.010)
%
% Source: Chatti & Zaabar (2012) NCHRP 720 Table 3-5.
% CI-to-IRI: calibrated to ORN 31 fair-condition thresholds.
%
% Reference values:
%   CI = 100 (new):          rrc_factor = 1.000
%   CI = 70 (good, mid):    rrc_factor = 1.028
%   CI = 55 (overlay trig.): rrc_factor = 1.042
%   CI = 40 (recon trig.):  rrc_factor = 1.056

beta = 0.014;
IRI = max(0, (100 - CI) / 15);
rrc_factor = 1 + beta * IRI;

end
```

B.2 config_base.m: Baseline Parameter Configuration

config_base.m assembles all model parameters into a single configuration structure `cfg`. This structure is passed as an argument to all other functions, ensuring that every simulation run operates from an identical, version-controlled parameter set. The values stored here correspond exactly to those reported in Appendix A.

```
function cfg = config_base()
% CONFIG_BASE Returns the baseline parameter structure for the
%           Nigerian pavement digital-twin-enabled simulation.
%
% cfg = config_base() creates a struct containing all model
% parameters: horizon, functional unit, traffic, climate,
% deterioration rates, maintenance thresholds, cost factors,
% carbon factors, and calibration targets.
%
% All monetary values are stored in NGN; conversion to GBP is
% applied in post-processing using cfg.exchange_rate.

% ANALYSIS HORIZON AND FUNCTIONAL UNIT
cfg.T           = 20;           % analysis horizon (years)
cfg.dt          = 1;           % time step (years)
cfg.road_length = 1000;        % functional unit length (m)
cfg.n_lanes     = 1;           % number of lanes
cfg.lane_width  = 3.65;        % lane width (m)
cfg.area        = cfg.road_length * cfg.lane_width; % m^2
cfg.base_year   = 2023;        % reference year

% EXCHANGE RATE
cfg.exchange_rate = 1024.4;      % NGN per GBP
cfg.inv_exchange_rate = 1/cfg.exchange_rate; % GBP per NGN

% DISCOUNT RATE
cfg.discount_rate = 0.12;        % annual discount rate (12%)

% TRAFFIC INPUTS
cfg.AADT_0       = 13667;        % two-way AADT (vehicles/day)
cfg.p_HGV        = 0.159;        % heavy vehicle share
cfg.traffic_growth = 0.04;        % annual growth rate
cfg.lane_split    = 0.5;         % directional split
cfg.ESA_per_HGV_noload = 1.6;    % ESAs per HGV, no overload
cfg.ESA_per_HGV_overload = 10.0; % ESAs per HGV, overloaded
cfg.loading_mode  = 'overload';  % active baseline mode
cfg.alpha         = 0.30;        % loading exponent
```

```

% Reference annual ESAs (calibration anchor)
daily_HGV = cfg.AADT_0 * cfg.p_HGV * cfg.lane_split;
cfg.ESA_ref = daily_HGV * 365 * cfg.ESA_per_HGV_noload;

% CLIMATE MULTIPLIER
cfg.k_climate = 1.15;           % humid tropical adjustment

% INITIAL CONDITION
cfg.CI_0 = 100;                % initial condition index
cfg.CI_failure = 40;           % reconstruction threshold
cfg.CI_cap = 100;              % upper bound

% DETERIORATION RATES (CI points per year, per band)
cfg.r_good = 2.2; % CI >= 70
cfg.r_mid = 3.0; % 55 <= CI < 70
cfg.r_poor = 4.2; % 40 <= CI < 55
cfg.r_vpoor = 6.0; % CI < 40

cfg.band_good = 70;
cfg.band_mid = 55;
cfg.band_poor = 40;

% MAINTENANCE THRESHOLDS
cfg.CI_minor = 65;
cfg.CI_overlay = 55;
cfg.CI_recon = 40;

% MAINTENANCE RECOVERY
cfg.dCI_minor = 4;
cfg.dCI_overlay = 28;

% INTERVENTION CONSTRAINTS
cfg.min_minor_interval = 3;
cfg.patch_area_frac = 0.08;
cfg.maintenance_delay = 0;

% COST PARAMETERS (NGN/m^2)
cfg.cost_asphalt_construction = 22500;
cfg.cost_asphalt_overlay = 9000;
cfg.cost_asphalt_minor_frac = 0.10;
cfg.cost_recon_ratio = 2.5;

cfg.cost_laterite_reseal_ratio = 0.60;
cfg.cost_laterite_construct_ratio = 0.75;

```

```

cfg.plastic_premium = 0.20;

cfg.cost_asphalt_minor    = cfg.cost_asphalt_overlay ...
                           * cfg.cost_asphalt_minor_frac;
cfg.cost_asphalt_recon    = cfg.cost_asphalt_overlay ...
                           * cfg.cost_recon_ratio;

cfg.cost_laterite_construction = cfg.cost_asphalt_construction ...
                                * cfg.cost_laterite_construct_ratio;
cfg.cost_laterite_reseal    = cfg.cost_asphalt_overlay ...
                              * cfg.cost_laterite_reseal_ratio;
cfg.cost_laterite_minor    = cfg.cost_laterite_reseal ...
                              * cfg.cost_asphalt_minor_frac;
cfg.cost_laterite_recon    = cfg.cost_laterite_construction;

cfg.cost_plastic_construction = cfg.cost_asphalt_construction ...
                                * (1 + cfg.plastic_premium);
cfg.cost_plastic_overlay    = cfg.cost_asphalt_overlay ...
                              * (1 + cfg.plastic_premium);
cfg.cost_plastic_minor    = cfg.cost_plastic_overlay ...
                              * cfg.cost_asphalt_minor_frac;
cfg.cost_plastic_recon    = cfg.cost_plastic_construction;

% CARBON PARAMETERS
cfg.ef_asphalt_mix    = 0.055;
cfg.ef_aggregate      = 0.010;
cfg.ef_bitumen        = 0.400;
cfg.density_asphalt   = 2400;
cfg.density_aggregate = 2000;
cfg.thick_overlay     = 0.050;
cfg.thick_patch       = 0.040;
cfg.thick_recon_AC    = 0.1262;
cfg.thick_recon_unbound = 0.35;
cfg.process_uplift    = 1.20;
cfg.seal_binder_rate  = 1.2;
cfg.seal_chip_rate    = 14.0;
cfg.plastic_co2_factor = 0.80;

% CALIBRATION TARGETS
cfg.target_asphalt_first_overlay = [8, 12];
cfg.target_asphalt_first_recon   = [15, 25];
cfg.target_laterite_first_reseal = [5, 9];
cfg.target_plastic_first_overlay = [10, 15];

```

```
% MATERIAL FACTOR LOOKUP
cfg.f_mat_asphalt = 1.00;
cfg.f_mat_laterite = 1.35;
cfg.f_mat_plastic = 0.75;
```

```
end
```

B.3 simulate_section.m: Annual Time-Step Loop

simulate_section.m is the main orchestration function. It initialises the state variables, calls deterioration_step.m and event_cost_emissions.m at each annual time step, accumulates costs and CO₂, and returns the complete time-series of outputs.

```
function out = simulate_section(material, cfg)
% SIMULATE_SECTION Runs the 20-year asset-level simulation for
% one pavement material.
```

```
T = cfg.T;
dt = cfg.dt;
```

```
CI = cfg.CI_0;
years_since_minor = cfg.min_minor_interval;
```

```
ci_series = zeros(T+1, 1);
cost_series = zeros(T+1, 1);
co2_series = zeros(T+1, 1);
cum_cost = zeros(T+1, 1);
cum_co2 = zeros(T+1, 1);
npv_series = zeros(T+1, 1);
```

```
intervention_log = {};
n_minor = 0;
n_overlay = 0;
n_recon = 0;
```

```
ci_series(1) = CI;
```

```
for t = 1:T
```

```
    ESA_t = compute_annual_ESA(t, cfg);
```

```
    [dCI, ~] = deterioration_step(CI, ESA_t, material, cfg);
    CI_post_det = max(0, CI - dCI);
```

```
    event_type = 'none';
```

```

years_since_minor = years_since_minor + dt;

if CI_post_det < cfg.CI_recon
    event_type = 'reconstruction';
elseif CI_post_det < cfg.CI_overlay
    event_type = 'overlay';
elseif CI_post_det < cfg.CI_minor ...
    && years_since_minor >= cfg.min_minor_interval
    event_type = 'minor';
end

[CI_new, cost_t, co2_t] = event_cost_emissions( ...
    CI_post_det, event_type, material, cfg);

if strcmp(event_type, 'minor')
    n_minor = n_minor + 1;
    years_since_minor = 0;
elseif strcmp(event_type, 'overlay')
    n_overlay = n_overlay + 1;
elseif strcmp(event_type, 'reconstruction')
    n_recon = n_recon + 1;
end

if ~strcmp(event_type, 'none')
    intervention_log{end+1} = struct( ...
        'year', t, 'type', event_type, ...
        'cost_NGN', cost_t, 'co2_t', co2_t);
end

CI = CI_new;

ci_series(t+1) = CI;
cost_series(t+1) = cost_t;
co2_series(t+1) = co2_t;
cum_cost(t+1) = cum_cost(t) + cost_t;
cum_co2(t+1) = cum_co2(t) + co2_t;
disc_factor = 1 / (1 + cfg.discount_rate)^t;
npv_series(t+1) = npv_series(t) + cost_t * disc_factor;

end

out.year = (0:T)';
out.CI = ci_series;
out.annual_cost_NGN = cost_series;

```

```

out.annual_co2_t      = co2_series;
out.cumulative_cost_NGN= cum_cost;
out.cumulative_co2_t  = cum_co2;
out.cumulative_cost_GBP= cum_cost * cfg.inv_exchange_rate;
out.npv_cost_GBP      = npv_series * cfg.inv_exchange_rate;
out.npv_total_GBP     = npv_series(end) * cfg.inv_exchange_rate;
out.total_co2_t       = cum_co2(end);
out.intervention_log  = intervention_log;
out.n_minor           = n_minor;
out.n_overlay         = n_overlay;
out.n_recon           = n_recon;
out.material          = material;

out.first_overlay_year = find_first_event( ...
    intervention_log, {'overlay','reconstruction'});

end

function ESA_t = compute_annual_ESA(t, cfg)
    if strcmp(cfg.loading_mode, 'overload')
        esa_per_HGV = cfg.ESA_per_HGV_overload;
    else
        esa_per_HGV = cfg.ESA_per_HGV_noload;
    end
    daily_HGV = cfg.AADT_0 * cfg.p_HGV * cfg.lane_split ...
        * (1 + cfg.traffic_growth)^(t-1);
    ESA_t = daily_HGV * 365 * esa_per_HGV;
end

function yr = find_first_event(log, types)
    yr = NaN;
    for k = 1:numel(log)
        if any(strcmp(log{k}.type, types))
            yr = log{k}.year;
            return;
        end
    end
end
end

```

B.4 deterioration_step.m: Annual Condition Loss

deterioration_step.m computes the condition-index loss for one annual time step, applying the piecewise band rates, traffic-loading factor, climate multiplier, and material-specific deterioration factor defined in Eq. (2).

```
function [dCI, f_traffic] = deterioration_step( ...
```

```

    CI, ESA_t, material, cfg)
% DETERIORATION_STEP Computes annual CI loss for one time step.

if CI >= cfg.band_good
    r_b = cfg.r_good;
elseif CI >= cfg.band_mid
    r_b = cfg.r_mid;
elseif CI >= cfg.band_poor
    r_b = cfg.r_poor;
else
    r_b = cfg.r_vpoor;
end

f_traffic = (ESA_t / cfg.ESA_ref)^cfg.alpha;

switch material
    case 'asphalt', f_mat = cfg.f_mat_asphalt;
    case 'laterite', f_mat = cfg.f_mat_laterite;
    case 'plastic', f_mat = cfg.f_mat_plastic;
    otherwise
        error('deterioration_step: unknown material "%s"', ...
            material);
end

dCI = r_b * f_traffic * cfg.k_climate * f_mat;
dCI = min(dCI, CI);

end

```

B.5 event_cost_emissions.m: Intervention Cost and CO₂

event_cost_emissions.m assigns agency cost (NGN) and embodied CO₂ (tonnes) to each maintenance event and returns the updated condition index after recovery is applied.

```

function [CI_new, cost_NGN, co2_t] = event_cost_emissions( ...
    CI, event_type, material, cfg)
% EVENT_COST_EMISSIONS Applies a maintenance event.

A = cfg.area;

switch event_type

    case 'none'
        CI_new = CI; cost_NGN = 0; co2_t = 0;

    case 'minor'

```



```

A_patch = A * cfg.patch_area_frac;
CI_new = min(cfg.CI_cap, CI + cfg.dCI_minor);
switch material
    case 'asphalt'
        cost_NGN = cfg.cost_asphalt_minor * A_patch;
        co2_t = asphalt_co2(cfg,A_patch,cfg.thick_patch,1.0);
    case 'laterite'
        cost_NGN = cfg.cost_laterite_minor * A_patch;
        co2_t = laterite_co2(cfg, A_patch, 'minor');
    case 'plastic'
        cost_NGN = cfg.cost_plastic_minor * A_patch;
        co2_t = asphalt_co2(cfg, A_patch, cfg.thick_patch, ...
            cfg.plastic_co2_factor);
end

case 'overlay'
    CI_new = min(cfg.CI_cap, CI + cfg.dCI_overlay);
    switch material
        case 'asphalt'
            cost_NGN = cfg.cost_asphalt_overlay * A;
            co2_t = asphalt_co2(cfg,A,cfg.thick_overlay,1.0);
        case 'laterite'
            cost_NGN = cfg.cost_laterite_reseal * A;
            co2_t = laterite_co2(cfg, A, 'reseal');
        case 'plastic'
            cost_NGN = cfg.cost_plastic_overlay * A;
            co2_t = asphalt_co2(cfg, A, cfg.thick_overlay, ...
                cfg.plastic_co2_factor);
    end

case 'reconstruction'
    CI_new = cfg.CI_0;
    switch material
        case 'asphalt'
            cost_NGN = cfg.cost_asphalt_recon * A;
            co2_t = asphalt_co2(cfg,A,cfg.thick_recon_AC,1.0) ...
                + unbound_co2(cfg, A, cfg.thick_recon_unbound);
        case 'laterite'
            cost_NGN = cfg.cost_laterite_recon * A;
            co2_t = laterite_co2(cfg, A, 'construct') ...
                + unbound_co2(cfg, A, cfg.thick_recon_unbound);
        case 'plastic'
            cost_NGN = cfg.cost_plastic_recon * A;
            co2_t = asphalt_co2(cfg, A, cfg.thick_recon_AC, ...

```

```

        cfg.plastic_co2_factor) ...
        + unbound_co2(cfg, A, cfg.thick_recon_unbound);
    end

    otherwise
        error('event_cost_emissions: unknown event "%s"', ...
            event_type);
    end
end
end

function co2_t = asphalt_co2(cfg, area, thickness, co2_factor)
    mass_AC = area * thickness * cfg.density_asphalt;
    co2_t = mass_AC * cfg.ef_asphalt_mix ...
        * cfg.process_uplift * co2_factor / 1000;
end

function co2_t = laterite_co2(cfg, area, mode)
    switch mode
        case {'reseal', 'minor'}
            co2_kg = cfg.seal_binder_rate * area * cfg.ef_bitumen ...
                + cfg.seal_chip_rate * area * cfg.ef_aggregate;
        case 'construct'
            co2_kg = area * 0.15 * cfg.density_aggregate ...
                * cfg.ef_aggregate;
        end
    co2_t = co2_kg * cfg.process_uplift / 1000;
end

function co2_t = unbound_co2(cfg, area, thickness)
    co2_t = area * thickness * cfg.density_aggregate ...
        * cfg.ef_aggregate * cfg.process_uplift / 1000;
end

```

C Vensim Model Equations and Coupling CSV

This appendix documents the final network-level Vensim model used in the study. It contains the governing stock-flow equations and the material-specific coupling parameters exported from MATLAB in `vensim_params.csv`.

C.1 Purpose of the Vensim Model

The Vensim model represents a stylised 100 km road network partitioned into material-specific acceptable-condition stocks and a shared backlog stock. Deterioration transfers road length from acceptable condition into backlog, while rehabilitation transfers road length back from backlog into acceptable condition subject to budget and policy constraints. Construction-phase (embodied) CO₂ and use-phase rolling-resistance CO₂ are accumulated as separate stocks and summed to yield whole-life CO₂ (see Section 3.3.4). The equations documented in this appendix reflect the baseline proportional-allocation structure; the priority-based allocation variant used for the robustness analysis in Section 4.3 was implemented in a separate MATLAB file (`time_to_crisis_priority.m`) that replaces the fixed s_m shares with dynamic priority weights $\pi_m = \dot{D}_m / \sum_{m'} \dot{D}_{m'}$ at each time step. The model is used to test budget, material-mix, and overload-enforcement scenarios over a 20-year horizon.

C.2 Full Vensim Equation Listing

Rehab capacity km=Annual Budget GBP / Weighted rehab Unit Cost
~ km/Year ~ |

Initial acceptable total km=Total network km-Initial backlog km
~ km ~ ~ :SUPPLEMENTARY |

Acceptable km asphalt= INTEG (Rehab Asphalt-Deterioration asphalt,
Initial acceptable km asphalt)
~ km ~ |

Acceptable KM Laterite= INTEG (Rehab Laterite-Deterioration laterite,
Initial acceptable km laterite)
~ km ~ |

Acceptable KM plastic= INTEG (Rehab Plastic-Deterioration Plastic,
Initial acceptable km plastic)
~ km ~ |

Annual Budget GBP=Budget Factor*Baseline annual budget GBP
~ GBP/Year ~ ~ :SUPPLEMENTARY |

Annual CO2 Addition t=(Rehab Asphalt*Unit rehab CO2 asphalt)
 +(Rehab Laterite*Unit rehab CO2 laterite)
 +(Rehab Plastic*Unit rehab CO2 plastic)
 ~ t/Year ~ |

Annual Cost Addition GBP=(Rehab Asphalt * Unit Rehab Cost asphalt)
 + (Rehab Laterite * Unit Rehab Cost laterite)
 + (Rehab Plastic * Unit Rehab Cost plastic)
 ~ GBP/Year ~ |

Backlog KM= INTEG (Total Deterioration-Total Rehab, Initial backlog km)
 ~ km ~ |

Backlog Fraction=Backlog KM/Total network km
 ~ Dmnl ~ :SUPPLEMENTARY |

Baseline annual budget GBP=844794 ~ GBP/Year ~ |
 Budget Factor=1 ~ Dmnl ~ |

Cumulative CO2 t= INTEG (Annual CO2 Addition t, 0)
 ~ t ~ :SUPPLEMENTARY |

Cumulative Cost GBP= INTEG (Annual Cost Addition GBP, 0)
 ~ GBP ~ :SUPPLEMENTARY |

Deterioration laterite=Acceptable KM Laterite
 / MAX(Effective TTL Laterite, 0.001)
 ~ km/Year ~ |

Deterioration Plastic=Acceptable KM plastic
 / MAX(Effective TTL Plastic, 0.001)
 ~ km/Year ~ |

Deterioration asphalt=Acceptable km asphalt
 / MAX(Effective TTL asphalt, 0.001)
 ~ km/Year ~ |

Unit rehab CO2 asphalt=23.5078 ~ t/km ~ |
 Unit rehab CO2 laterite=3.82907 ~ t/km ~ |
 Unit rehab CO2 plastic=24.771 ~ t/km ~ |
 Unit Rehab Cost asphalt=21256.9 ~ GBP/km ~ |
 Unit Rehab Cost laterite=11820.8 ~ GBP/km ~ |
 Unit Rehab Cost plastic=26067.1 ~ GBP/km ~ |

Weighted rehab Unit Cost=((Rehab share asphalt*Unit Rehab Cost asphalt)
 +(Rehab share laterite*Unit Rehab Cost laterite)
 +Rehab share plastic*Unit Rehab Cost plastic)
 ~ GBP/km ~ :SUPPLEMENTARY |

Rehab response time=1
 ~ Year ~ |

Rehab Plastic=min((Backlog KM*Rehab share plastic)/Rehab response time,
 Rehab capacity km*Rehab share plastic)
 ~ km/Year ~ |

Total Rehab=Rehab Asphalt+Rehab Laterite+Rehab Plastic
 ~ km/Year ~ |

Rehab Asphalt=min((Backlog KM*Rehab share asphalt)/Rehab response time,
 Rehab capacity km*Rehab share asphalt)
 ~ km/Year ~ |

Total Deterioration=Deterioration laterite
 +Deterioration Plastic+Deterioration asphalt
 ~ km/Year ~ |

Rehab Laterite=min((Backlog KM*Rehab share laterite)/Rehab response time,
 Rehab capacity km*Rehab share laterite)
 ~ km/Year ~ |

Total acceptable km=Acceptable km asphalt
 +Acceptable KM Laterite+Acceptable KM plastic
 ~ km ~ |

Initial acceptable km plastic=8.5 ~ km ~ |
 Initial acceptable km laterite=17 ~ km ~ |
 Initial backlog km=15 ~ km ~ |
 Initial acceptable km asphalt=59.5 ~ km ~ |

Acceptable fraction=Total acceptable km / Total network km
 ~ Dmn1 ~ :SUPPLEMENTARY |

Effective TTL asphalt=TTL asphalt/Overload damage multiplier
 ~ Year ~ |

Effective TTL Laterite=TTL laterite/Overload damage multiplier
 ~ Year ~ |

Effective TTL Plastic=TTL Plastic/Overload damage multiplier

```

~ Year ~ |

Overload damage multiplier=1 ~ Dmnl ~ |
Rehab share asphalt=0.7 ~ Dmnl ~ |
Rehab share laterite=0.2 ~ Dmnl ~ |
Rehab share plastic=0.1 ~ Dmnl ~ |
Total network km=100 ~ km ~ |
TTL asphalt=5 ~ Year ~ |
TTL laterite=3.33333 ~ Year ~ |
TTL Plastic=7 ~ Year ~ |

FINAL TIME = 20 ~ Year ~ |
INITIAL TIME = 0 ~ Year ~ |

```

C.3 Additional Equations: Use-Phase CO₂ and Priority Allocation

Use-phase CO₂ stocks (added in the updated model)

The use-phase rolling-resistance CO₂ stock accumulates annually from the acceptable-condition kilometres of each material. The governing equations in Vensim notation are:

```

Annual use-phase CO2 asphalt = Acceptable asphalt km
    * Unit_up_CO2_asphalt * Traffic_growth_factor
    ~ t/Year ~

```

```

Annual use-phase CO2 laterite = Acceptable laterite km
    * Unit_up_CO2_laterite * Traffic_growth_factor
    ~ t/Year ~

```

```

Annual use-phase CO2 plastic = Acceptable plastic km
    * Unit_up_CO2_plastic * Traffic_growth_factor
    ~ t/Year ~

```

```

Traffic_growth_factor = (1 + traffic_growth_rate)^Time ~ Dmnl ~
Unit_up_CO2_asphalt = 4.5427 ~ t/km/Year ~
Unit_up_CO2_laterite = 6.2462 ~ t/km/Year ~
Unit_up_CO2_plastic = 3.9749 ~ t/km/Year ~
traffic_growth_rate = 0.04 ~ 1/Year ~

```

```

Cum use-phase CO2 = INTEG(
    Annual use-phase CO2 asphalt + Annual use-phase CO2 laterite
    + Annual use-phase CO2 plastic, 0) ~ t ~

```

```

Cum whole-life CO2 = Cum embodied CO2 + Cum use-phase CO2 ~ t ~

```

Backlog km are excluded from use-phase accumulation (see Section 3.3.4 for justification).

Priority-based rehabilitation allocation (sensitivity analysis)

In the priority-based variant (`time_to_crisis_priority.m`), the fixed rehabilitation shares s_m are replaced by dynamic priority weights:

$$\pi_m(t) = \frac{A_m(t)/\tau_m}{\sum_{m'} A_{m'}(t)/\tau_{m'}} \quad (1)$$

These weights are recomputed at each annual time step. All other equations are identical to the baseline proportional-allocation structure documented above.

C.4 Exported MATLAB-to-Vensim Coupling CSV

Table C.1: Material-specific parameters exported to `vensim_params.csv`.

Key	Material	AADT	% HGV	ESA/HV	Growth	Discount	Life
asphalt	Asphalt (conventional)	13,667	0.159	10.0	0.04	0.12	20
laterite	Laterite (sealed base + reseal)	13,667	0.159	10.0	0.04	0.12	20
plastic	Plastic-modified asphalt	13,667	0.159	10.0	0.04	0.12	20

Table C.2: Maintenance interval and annualised parameters from `vensim_params.csv`.

Key	Mean maint. int.	Mean minor int.	Mean overlay int.	Cost per km-year (£)	CO ₂ per km-year (t)
asphalt	2.4	5.0	5.0	8857.06	9.7949
laterite	1.7143	3.6667	3.3333	6895.49	2.2336
plastic	3.0	7.0	7.0	8689.02	6.6056

Table C.3: Lifetime cost, carbon, and intervention counts from `vensim_params.csv`.

Key	NPV cost (£)	Construction- phase CO ₂ (t)	Loading mode	n_{minor}	n_{overlay}	n_{recon}
asphalt	99872.50	195.8981	overload	3	3	0
laterite	79687.93	44.6725	overload	4	4	0
plastic	110691.65	132.1120	overload	2	2	0

C.5 Commentary

The equation listing confirms that the network model is built around three material-specific acceptable-condition stocks, a shared backlog stock, and budget-constrained rehabilitation flows. The coupling CSV confirms that differences in mean overlay interval, annualised cost intensity, annualised CO₂ intensity, and intervention frequency are transferred directly from MATLAB into Vensim, enabling asset-level material behaviour to influence network-level backlog, cost, and emissions outcomes in a traceable and reproducible form.

D Supplementary Sensitivity and Monte Carlo Plots

D.1 Initial Backlog Sensitivity

Table D.1: Initial backlog sensitivity: year-20 backlog (km) for S0, S1, S2 and selected scenarios under varying initial backlog fractions. S6 outperforms S5 and S4 is worse than S0 at all levels tested.

Init. bklg (%)	Init. bklg (km)	S0 yr-20	S1 yr-20	S2 yr-20	S4 yr-20	S5 yr-20	S6 yr-20
5%	5	17.054	17.054	17.054	19.243	12.597	10.698
15%	15	17.054	17.054	17.054	19.244	12.597	10.683
25%	25	17.055	17.055	17.055	19.245	12.597	10.668
35%	35	17.055	17.055	17.055	19.246	12.598	10.653

This appendix provides supplementary plots supporting the uncertainty and sensitivity analysis reported in Chapter 4.

D.2 One-at-a-Time Sensitivity Analysis

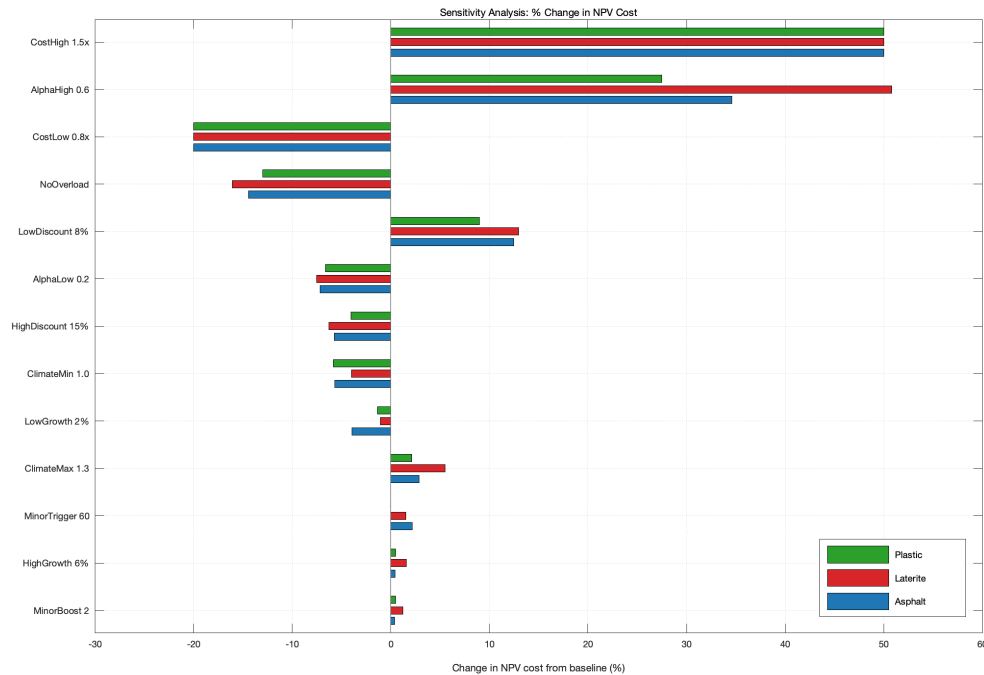


Figure D.1: Supplementary one-at-a-time sensitivity tornado chart showing percentage change in 20-year NPV cost relative to baseline for the tested parameter variations across the three pavement materials.

D.3 Monte Carlo Box Plots

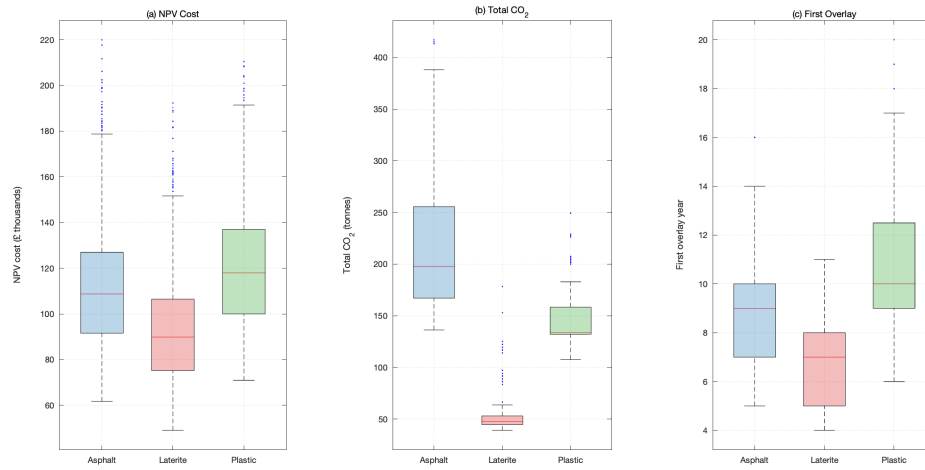


Figure D.2: Supplementary Monte Carlo box plots ($N = 2,000$) for 20-year NPV cost, construction-phase CO₂, and first overlay year. Bounds reflect six sampled parameters; ODM, initial backlog fraction, and rehabilitation shares were not varied at the asset level. ODM and initial backlog are varied in the network-level Monte Carlo (Table 4.10).

D.4 Appendix Summary

The tornado chart shows that model outputs are most sensitive to cost assumptions and loading severity. The Monte Carlo box plots confirm that sealed laterite retains a strong CO₂ advantage and a strong though not absolute cost advantage within the sampled parameter space.

E Structural Adequacy Check

E.1 Purpose and Scope

This appendix provides a structural adequacy check for the pavement cross-section adopted throughout the modelling. The purpose is not to execute a full pavement design for a specific Nigerian corridor, but to confirm that the modelled section is broadly consistent with a defensible structural basis under the traffic conditions assumed in the study. The check follows the methodology of TRL Overseas Road Note 31 (Transport Research Laboratory, 1993). The comparison used here is a plausibility screen rather than a rigorous layer-equivalence proof.

E.2 Design Traffic Estimate

Table E.1: Design traffic estimate for structural adequacy check.

Parameter	Value	Notes
Two-way AADT	13,667 veh/day	Baseline model input
Heavy vehicle share	15.9%	Baseline model input
Lane split factor	0.50	Single-lane functional unit
Design-lane daily HGVs	1,087 HGV/day	$AADT \times 0.159 \times 0.50$
ESA/HGV (overload baseline)	10.0	Documented axle-load exceedance
ESA/HGV (no overload)	1.6	Non-overloaded reference
Annual traffic growth	4%/yr	Baseline model input
Annual ESA at year 0 (overload)	3,969 k ESA/yr	$1087 \times 365 \times 10.0$
Annual ESA at year 0 (no overload)	635 k ESA/yr	$1087 \times 365 \times 1.6$
20-year cumulative ESA (overload, 4% growth)	117.8 M ESA	$\sum_{t=0}^{19} ESA_0 \times (1.04)^t$
20-year cumulative ESA (no overload, 4% growth)	18.8 M ESA	As above with ESA/HGV = 1.6
ORN 31 traffic class (overload baseline)	T5 / T6	30–300 M ESA range
ORN 31 traffic class (no overload)	T4	10–30 M ESA range

Under the baseline overloaded condition, 20-year cumulative design traffic is 117.8 M ESA, placing the section at the T5/T6 boundary. Under the no-overload reference, cumulative traffic falls to 18.8 M ESA (class T4). This pronounced sensitivity to the ESA/HGV factor underlines the structural significance of the enforcement scenarios in Chapter 4.

E.3 Thickness Comparison Against ORN 31

A CBR of 5% is adopted as the representative subgrade value, consistent with weak lateritic subgrades reported in the Nigerian case-study literature reviewed in Chapter 2.

Table E.2: Thickness comparison: modelled section vs ORN 31 reference (CBR = 5%, T5 traffic class).

Layer	Modelled (mm)	ORN 31 ref (mm)	Margin (mm)	Assessment
Surface course	50	40	+10	Broadly adequate ✓
Roadbase / granular base	—	150	—	Within unbound layer
Unbound layer (combined)	350	200	+150	Broadly adequate ✓
Total structural depth	400	390	+10	Broadly adequate ✓

Notes: ORN 31 reference for T5/CBR 5%: 40 mm asphaltic surface + 150 mm crushed-stone road-base + 200 mm granular subbase = 390 mm total. The modelled structural depth of 400 mm exceeds the ORN 31 reference by 10 mm. This is a plausibility screen, not a rigorous layer-equivalence proof; it supports the section's use as a broadly credible engineering basis.

E.4 ESA Sensitivity Check

Table E.3: ESA sensitivity check: 20-year cumulative traffic and ORN 31 class under alternative assumptions.

Case	Growth rate	ESA/HGV	20-yr cumulative (M ESA)	ORN 31 class	Adequate?
Low-load, low growth	2%/yr	1.6	13.0	T4	✓
Low-load, baseline growth	4%/yr	1.6	18.8	T4	✓
Low-load, high growth	6%/yr	1.6	27.9	T4/T5	✓
Overload, low growth	2%/yr	10.0	81.3	T5	✓
Overload, baseline growth	4%/yr	10.0	117.8	T5/T6	✓
Overload, high growth	6%/yr	10.0	174.3	T6	✓

Across all six cases the modelled section is broadly adequate against the ORN 31 reference depth. The section operates near the T5/T6 boundary under the baseline overloaded condition, consistent with the accelerated deterioration observed in the asset-level results.

E.5 Conclusion

The modelled pavement cross-section is broadly consistent with ORN 31 structural requirements across the full range of traffic-loading assumptions tested in this study. This plausibility screen supports its use as a credible engineering basis rather than a purely convenient abstraction. The primary structural risk is the sensitivity of design-class assignment to the ESA/HGV factor, which shifts from T4 (no overload) to T5/T6 (baseline overload), reinforcing the policy significance of the overload-enforcement scenarios in Chapter 4.